



Project Report On

CAR RENTAL MANAGEMENT SYSTEM

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1. PROJECT DESCRIPTION

1.1 PROJECT PROFILE

Project Name:	Car Rental Management System
Operating System:	Windows 10,11
Server-Side Scripting language:	PHP Version 7.4.26
Front End:	HTML, Bootstrap 5, CSS, jQuery, Sweet alert, Data Table, Font Awesome
Back End:	MY SQL
Prepared By:	Dhruv Rana [21B123]
Submitted to:	M K Institute of Computer Studies, Bharuch.
Guide:	Prof. Ms. Nidhi Solanki

1.2 PROJECT INTRODUCTION

This **Car Rental System** project is designed to aid the car rental company to enable renting of cars through an online system. It helps the users to search for available cars view profile and book the cars for the time period. It has a user-friendly interface which helps the user to check for cars and rent them for the period specified. They could also make payment online. The rental cars shall be categorized into economy, premium etc. Based on the type of car required by the customer, the user shall be able to make bookings. The use of internet technology has made it easy for the customers to rent a car any time. This Car Rental System makes the bookings easy. It saves time and labour. The tool shall ask the user for information such as the date and time of journey, type of car etc. Also, it will need an identification number. Using these details, the tool shall help the customer to book a car for the journey.

1.3 PROJECT DEFINITION

A Car Rental Management System (CRMS) is a software solution designed to streamline and automate the operations of a car rental business. This type of system is used by car rental companies to manage their fleet, reservations, bookings, customer information, billing, and other related tasks. Here are some key features and components typically found in a Car Rental Management System:

1. **Fleet Management:** Allows the management of the entire fleet of vehicles, including details such as make, model, year, registration information, and current status (available, rented, in maintenance).
2. **Reservation System:** Enables customers to make reservations for specific vehicles, dates, and times. Provides a calendar view for easy tracking of reservations.
3. **Customer Database:** Stores customer information, including contact details, address, rental history, etc. Facilitates the creation of customer profiles for faster and more efficient transactions.
4. **Booking and Scheduling:** Manages the scheduling of vehicle pickups and returns. Sends notifications to customers about their upcoming reservations and provides information on pickup/drop-off locations.
5. **Reporting and Analytics:** Provides reports on key metrics such as revenue, utilization rates, and maintenance costs. Enables data analysis to optimize fleet management and business operations.
6. **Maintenance Tracking:** Monitors the maintenance schedules of each vehicle. Alerts staff when a vehicle is due for maintenance or repairs.
7. **User Roles and Permissions:** Defines different user roles (admin, staff, customer) with varying levels of access and permissions.
8. **Mobile Accessibility:** Provides a mobile-friendly interface for customers to make reservations and access their accounts.
9. **Insurance Management:** Manages insurance information for each vehicle and ensures that customers are adequately covered during the rental period.

2. PROJECT PLANNING

2.1 SCOPE

- The car rental system to keep detail records of both the cars and the customers, the duration they rent car as well as the type of car they rent.
- The system will be mainly design for small a company that renders it car rental services to customers.
- The system will have the ability to generate and print invoice for each successful transaction.
- Level of Access
- The system will have two levels of access
- The administrator
- Customer
- It satisfies the user requirement.
- Have a good use of interface.
- Easy to understand by the user and operator.

2.3 GOAL

The primary goal of a car rental management system is to facilitate efficient and seamless management of all aspects of the car rental process. This includes everything from customer reservations to vehicle maintenance. The specific goals of a car rental management system typically include:

1. **Optimizing Fleet Utilization:** Ensure that vehicles are utilized effectively by matching customer demand with available inventory. This minimizes idle time for vehicles and maximizes revenue for the rental company.
2. **Streamlining Reservation Process:** Enable customers to easily make reservations through various channels such as online platforms, mobile apps, or in-person at rental offices. Provide real-time availability information and confirmation of bookings.
3. **Enhancing Customer Experience:** Provide a user-friendly interface for customers to browse available vehicles, select options, and complete bookings. Offer convenient features such as online check-in, digital contracts, and mobile key access to improve the rental experience.

4. Ensuring Fleet Maintenance and Safety: Implement systems to track vehicle maintenance schedules, service history, and safety inspections. Promptly address any maintenance issues to ensure that rental vehicles are safe and in good working condition.
5. Managing Customer Relationships: Maintain a database of customer profiles, preferences, and rental history to provide personalized service. Offer loyalty programs, promotions, and discounts to incentivize repeat business.
6. Improving Operational Efficiency: Automate routine tasks such as vehicle assignment, inventory management, and reporting. Reduce manual errors and administrative overhead to improve overall efficiency.
7. Providing Insights for Decision Making: Generate reports and analytics on various aspects of rental operations, such as revenue, utilization rates, and customer demographics. Use these insights to make informed decisions and optimize business performance.

2.3 OBJECTIVE

To produce a web-based system that allow customer to register and Booking car online and for the company to effectively manage their car rental business.

To ease customer's task whenever they need to rent a car

2.4 CONSTRAINT

A car rental management system may encounter various constraints or limitations, which can affect its functionality, performance, and effectiveness.:

1. Resource Limitations: Constraints related to hardware resources (such as servers, storage) or software resources (such as memory, processing power) can limit the scalability and performance of the system.
2. Geographical Constraints: Operational constraints related to geographic locations, such as limited coverage areas for vehicle delivery or pickup services, may affect the accessibility and availability of the rental service.
3. Security Constraints: Concerns about data security, confidentiality, and fraud prevention may impose constraints on access controls, encryption methods, and authentication mechanisms implemented within the system.

4. Customer Constraints: Customer preferences, behaviours, and expectations may impose constraints on the system's features, usability, and service offerings, requiring flexibility and adaptability to meet diverse needs.
5. Environmental Constraints: Factors such as weather conditions, traffic patterns, and seasonal fluctuations in demand may impose constraints on vehicle availability, maintenance schedules, and operational logistics.

3. ENVIRONMENT DESCRIPTION

3.1 HARDWARE SPECIFICATION

- Operating System: Windows 10,11
- Browser Support: Google Chrome, Microsoft Edge
- Minimum Ram: 4 GB
- Hard Disk:100 Mb
- Processor: Intel(R) Core (TM) i3-1005G1 CPU @ 1.20GHz 1.19 GHz

3.2 TECHNOLOGY USED/ SOFTWARE REQUIREMENTS

- PHP (Hypertext Pre-processor)
- MySQL 5.7.36
- PhpMyAdmin 5.1.1
- WampServer 3.2.6 - 64bit

4. SYSTEM DESCRIPTION

4.1 Feasibility Study

Preliminary investigation examines project feasibility, the likelihood the system will be useful to the organization. The main objective of the feasibility study is to test the Technical, Operational and Economical feasibility for adding new modules and debugging old running system. All system is feasible if they are unlimited resources and infinite time. There are aspects in the feasibility study portion of the preliminary investigation:

➤ **Technical Feasibility**

➤ **Operation Feasibility**

➤ **Economical Feasibility**

Technical Feasibility

The technical issue usually raised during the feasibility stage of the investigation includes the following:

Do the proposed equipments have the technical capacity to hold the data required to use the new system?

Are there technical guarantees of accuracy, reliability, ease of access and data security?

Operational Feasibility

Proposed projects are beneficial only if they can be turned out into information system. That will meet the organization's operating requirements. Operational feasibility aspects of the project are to be taken as an important part of the project implementation.

Economical Feasibility

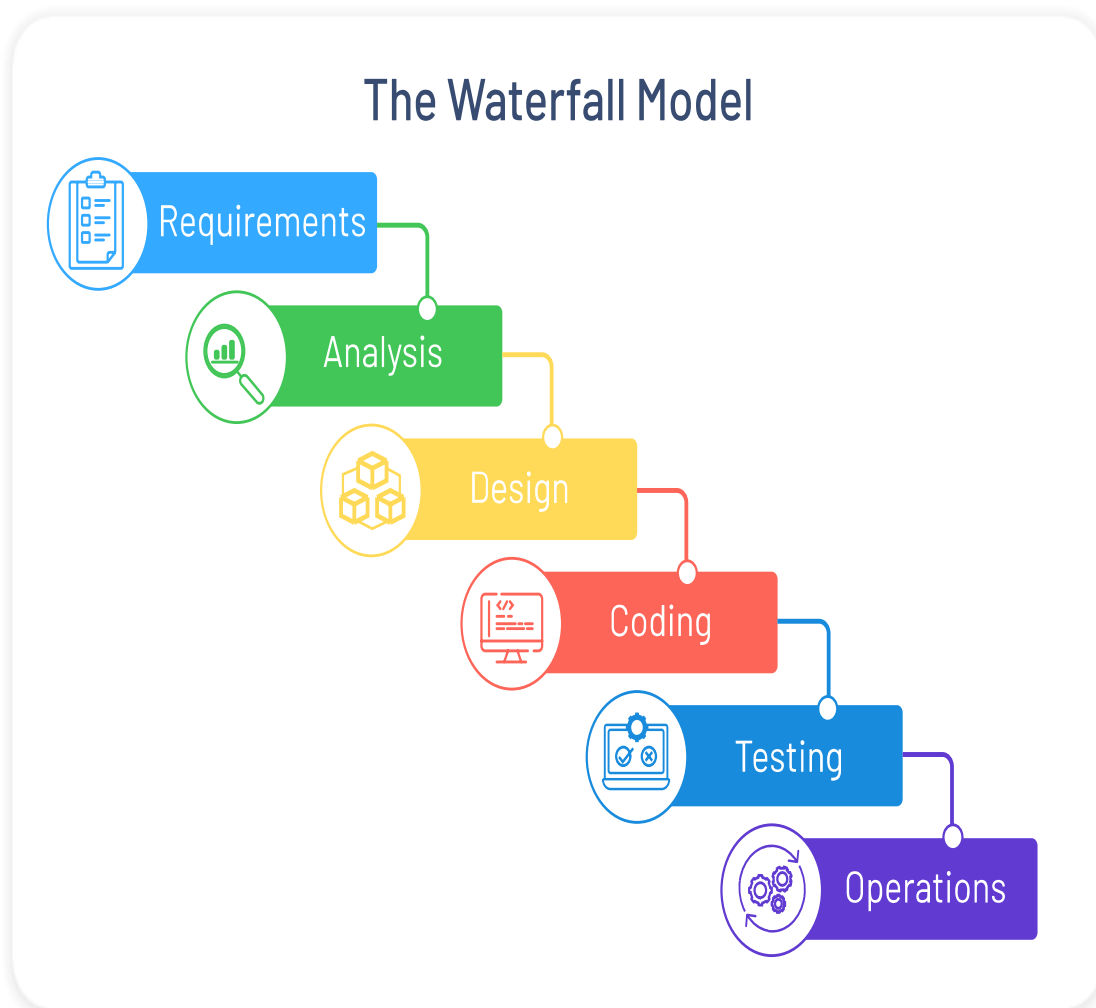
A system can be developed technically and that will be used if installed must still be a good investment for the organization. In the economical feasibility, the development cost in creating the system is evaluated against the ultimate benefit derived from the new systems. Financial benefits must equal or exceed the costs.

4.2 Requirement Analysis/ Data Gathering

1. **Stakeholder Identification:** Identify key stakeholders such as business owners, managers, rental agents, and customers. Understand their roles, responsibilities, and expectations regarding the car rental system.
2. **Business Objectives:** Define the primary goals of the car rental business, such as increasing revenue, expanding customer base, or improving operational efficiency.
3. **User Stories and Use Cases:** Collect user stories and use cases to understand how different users will interact with the system. Identify typical scenarios, such as making a reservation, renting a car, returning a car, and handling customer inquiries.
4. **Functional Requirements:** Identify and document the core functionalities of the car rental system, including:
 - User registration and authentication
 - Car inventory management
 - Reservation and booking system
 - Rental agreements and terms
 - Pricing and billing
 - Vehicle return and inspection process
5. **Non-functional Requirements:** Gather non-functional requirements, such as performance, security, and scalability expectations. Consider factors like system response time, data security, and the ability to handle a varying number of users.
6. **Integration Requirements:** Identify any external systems that need to be integrated, such as payment gateways, GPS tracking, or customer relationship management (CRM) systems.
7. **Regulatory and Compliance Requirements:** Understand and document any legal or regulatory requirements related to the car rental industry. Ensure that the system complies with data protection and privacy laws.
8. **Data Requirements:** Define the data entities, such as customer information, vehicle details, reservations, and rental history. Determine the data storage and retrieval mechanisms, including database requirements.

9. User Interface (UI) and User Experience (UX) Requirements: Specify the design and usability expectations for the user interface. Consider factors such as ease of navigation, mobile responsiveness, and accessibility.
10. Reporting and Analytics: Determine the reporting and analytics features needed by the business. Identify key performance indicators (KPIs) and reporting tools for monitoring the success of the car rental service.
11. Testing and Quality Assurance: Define the testing requirements, including functional testing, security testing, and user acceptance testing. Establish quality assurance processes to ensure the reliability and performance of the system.
12. Training and Support: Outline training requirements for users and support staff. Consider the need for customer support features, such as live chat or a helpdesk.
13. Feedback Mechanisms: Establish mechanisms for collecting feedback from users and stakeholders to continuously improve the system.
14. Prototyping and Validation: Create prototypes or mock ups to validate the proposed system with stakeholders before development begins.

4.3 System development process model



- The waterfall model is a sequential design process, often used in software development processes, in which progress is seen as flowing steadily downwards (like a waterfall) through the phases of Conception, Initiation, Analysis, Design, Construction, Testing, Production/Implementation and Maintenance.
- In Royce's original waterfall model, the following phases are followed in order:
 - Requirement's specification
 - Design
 - Construction (Implementation or coding)

- Integration
 - Testing and debugging (Validation)
 - Installation
 - Maintenance
-
- Thus, the waterfall model maintains that one should move to a phase only when its proceeding phase is completed and perfected. However, there are various modified waterfall models (including Royce's final model) that may include slight or major variations on this process.
-
- Stages of the Waterfall Model Explained: -

• Requirement Analysis and Definition

All possible requirements of the system to be developed are captured in this phase. Requirements are a set of functions and constraints that the end user (who will be using the system) expects from the system. The requirements are gathered from the end user at the start of the software development phase. These requirements are analysed for their validity, and the possibility of incorporating the requirements in the system to be developed is also studied. Finally, a requirement specification document is created which serves the purpose of guideline for the next phase of the model

• System and Software Design

Before starting the actual coding phase, it is highly important to understand the requirements of the end user and also have an idea of how the end product should look like. The requirement specifications from the first phase are studied in this phase and a system design is prepared. System design helps in specifying hardware and system requirements and also helps in defining the overall system architecture. The system design specifications serve as an input for the next phase of the model.

• Implementation and Unit Testing

On receiving system design documents, the work is divided in modules/units and actual coding is started. The system is first developed in small programs called units, which are integrated in the next phase. Each unit is developed and tested for its functionality; this is referred to as unit testing. Unit testing verifies if the modules/units meet their specifications.

• Integration and System Testing

As specified above, the system is first divided into units which are developed and tested for their functions. These units are integrated into a complete system during integration phase and tested to check if all modules/units coordinate with each

other, and the system as a whole behaves as per the specifications. After successfully testing the software, it is delivered to the customer.

• **Operations & Maintenance**

This phase of the waterfall model is virtually a never-ending phase. Generally, problems with the system developed (which are not found during the development life cycle) come up after its practical use starts, so the issues related to the system are solved after deployment of the system. Not all the problems come into picture directly, but they arise from time to time and need to be solved; hence this process is referred to as maintenance.

Advantages of the Waterfall Model: -

- There is clear compartmentalization of work and control in the waterfall model. Since there is compartmentalization, it is easier to set schedule for the tasks to be completed within a specified time period.
- The other advantage of this model is that only after the work for a particular phase is over, does the other phase start, due to which there is no overlapping of phases or the product does not have to go through different iterative steps.
- This model is the easiest to implement in the eyes of most of the managers, due to its linear model. Since the processes of this model are carried out in linear manner, the cost of resources is reduced to a large extent, which in turn helps in reducing the cost of the project considerably.
- Lastly, the documentation and testing happen at the end of each phase, which helps in maintaining the quality of the project.

5.ADVANTAGES OF SYSTEM

5.1 ADVANTAGES OF SYSTEM

Easy and instant booking: This is one of the most obvious advantages of online car rental system development. Customers will be able to book a vehicle from wherever they are, at any time, thanks to the car rental system development. As a result, this convenience will improve your customers' experience, and as a result, increase your revenue.

Effective Fleet Management: A car rental fleet management system makes it easy for business owners to manage all their vehicles in one place. The business owner can add or remove vehicles from the fleet and check the status of live bookings status.

Automatic Fare Calculation: Based on the date, time, and vehicle type, the online car rental software program automatically calculates and provides the fare to customers. The software can be set to calculate fares using different parameters set by the business owner. By using this booking system, businesses can avoid some issues pertaining to the traditional booking system, such as price negotiations, and reduce the risk of miscommunication with their customers.

Be Informed with The Customer and Driver Information: Car rental online booking system stores the information of the driver and customer and makes this information available to business owners. A business owner can also permanently delete the account of a customer or user if necessary.

Billing Made Easy: Online car rental businesses face many challenges, including generating invoices for customers. A car rental system development company can simplify this time-consuming process to a great extent. Car rental software can generate the invoice itself as soon as the trip ends by calculating the fuel consumption, km driven, rented time, and so many other things.

Round the Clock Availability: One of the greatest benefits of an online car rental booking system is its availability 24 hours a day, 7 days a week. Offering round-the-clock support will boost the company's revenue by enticing more customers.

Track the Car Availability: Even after the customer confirms the booking, it is still a long process to confirm the availability of the vehicle. Sometimes the company might fail to fulfil the needs of the customers. A car rental booking software program will provide the customer with a list of all the available cars during their specific date and time.

5.2 Security Issues Add Precaution

1. **User Authentication and Authorization:** Implement a robust authentication mechanism to ensure that only authorized users can access the system. Utilize multi-factor authentication (MFA) where possible to add an extra layer of security.
2. **Secure Communication:** Ensure that all communication between the client and the server is encrypted using protocols such as HTTPS. This helps prevent eavesdropping and data tampering.
3. **Data Encryption:** Encrypt sensitive data such as user credentials, payment information, and personal details to protect them from unauthorized access in case of a data breach.
4. **Secure Payment Processing:** Use reputable payment gateways and follow industry standards for handling payment information securely. Avoid storing payment card details on your servers and instead utilize tokenization or other secure methods.
5. **Regular Security Audits and Updates:** Conduct regular security audits of the system to identify vulnerabilities and weaknesses. Keep all software components and libraries up to date with the latest security patches and updates.
6. **Monitoring and Logging:** Implement logging mechanisms to track user activities and system events. Set up alerts for suspicious activities or anomalies that could indicate a security breach. Monitoring can help detect and respond to security incidents in a timely manner.
7. **Secure Password Handling:** Enforce strong password policies, such as minimum length, complexity requirements, and expiration periods. Encourage users to use unique passwords and consider implementing password hashing and salting techniques to securely store passwords.
8. **Secure File Uploads:** If your system allows users to upload files (e.g., upload car images, etc), implement measures to prevent malicious file uploads. Use file type verification, size limits, and scanning for malware to mitigate risks associated with file uploads.
9. **Physical Security Measures:** If applicable, ensure physical security measures are in place to protect hardware components such as servers and databases from unauthorized access or tampering.

6. SYSTEM DESIGNING

What is Data Dictionary?

In database management systems, a file that defines the basic organization of a database. A data dictionary contains a list of all files in the database, the number of records in each file, and the names and types of each field. Most database management systems keep the data dictionary hidden from users to prevent them from accidentally destroying its contents.

Data dictionaries do not contain any actual data from the database, only book keeping information for managing it. Without a data dictionary, however, a database management system cannot access data from the database.

A data dictionary is a collection of descriptions of the data objects or items in a data model for the benefit of programmers and others who need to refer to them. A first step in analyzing a system of objects with which users interact is to identify each object and its relationship to other objects. This process is called data modelling and results in a picture of object relationships. After each data object or item is given a descriptive name, its relationship is described (or it becomes part of some structure that implicitly describes relationship), the type of data (such as text or image or binary value) is described, possible predefined values are listed, and a brief textual description is provided. This collection can be organized for reference into a book called a data dictionary.

Why is a Data Dictionary Important?

Analysts use data dictionary for the following reasons:

- To manage the details in the large system.
- To communicate a common meaning for all system elements.
- To document the features of the system.
- To locate error and omission in the system.

6.1 DATA DICTIONARY

Data Element	Data Type	Size	Constraint	Storage Table	Description
Admin_id	Int		Primary key	Admin Master	Used for unique identifier for each admin system / unique identifier for each event
Username	Varchar	100	Not Null	Admin Master	The unique identifier or username associated with an administrative account
Password	Varchar	100	Not Null	User Master, Admin Master	It display password
Updation_Date	Timestamp		Not Null	Booking Master, User Master, Admin Master, Brand Master, Vehicle Master	The date when the rental record was last updated
Id	Int		Not Null	Booking, User, Admin, Brands, ContactUsInfo, ContactUsQuery, Pages, Vehicles, Testimonial	Unique identifier for each customer
Booking number	Big int		Not Null	Booking Master	It stores pick up date
User email	Varchar	100	Not Null	Booking Master, Testimonial	Email address of the customer
Vehicle Id	Int		Primary key	Vehicle Master	It stores pick up location
From Date	Varchar	20	Not Null	Booking Master	It stores pick up date
To Date	Varchar	20	Not Null	Booking Master	It stores from up date
Message	Varchar	255	Not Null	Booking Master, ContactUsQuery	It stores additional accessories of the user
Status	Int		Not Null	Booking Master, ContactUsQuery, Testimonial	State where the customer resides

Posting Date	Timestamp		Not Null	Booking Master, ContactUsQuery, ContactUsInfo, Testimonial	The unique identifier or username associated with an administrative account
LastUpdation Date	Timestamp		Not Null	Booking Master	The most recent date when any changes were made to the user's information or the rental details
Brands Id	Int		Primary key	Brand Master	A unique identifier for each car in the rental system
Brand name	Varchar	120	Not Null	Brand Master	The name of the car rental brand or company.
Creation Date	Timestamp		Not Null	Brand Master	The date when the car rental brand or company was established or created
Contact info Id	Int		Primary key	ContactUsInfo Master	A unique identifier for each car in the rental system
Address	Tiny text		Not Null	User Master, ContactUsInfo	The physical address of the car rental office or headquarters
Email Id	Varchar	255	Not Null	User Master, ContactUsInfo, ContactUsQuery	The email address that customers can use to contact the car rental service
Contact No	Char	11	Not Null	User Master, ContactUsInfo, ContactUsQuery	The phone number that customer can use to reach the car rental service
Contact query Id	Int		Primary key	ContactUsQuery Master	A unique identifier for each car in the rental system
Name	Varchar	100	Not Null	ContactUsQuery	The physical address of the car rental office or headquarters
Page Id	Int		Primary key	Page	A unique identifier for each car in the rental system
Page Name	Varchar	255	Not Null	Page	The physical address of the car rental office or headquarters
Type	Varchar	255	Not Null	Page	The email address that customers can use to contact the car rental service
Detail	Long text		Not Null	Page	The phone number those customers can use to reach the car rental service
Subscription Id	Int		Primary key	Subscription	A unique identifier for each subscription record
Subscriber Email	Varchar	120	Not Null	Subscription	The email address of the subscriber
Testimonial Id	Int		Primary key	Testimonial	A unique identifier for each testimonial
Testimonial	Medium text		Not Null	Testimonial	The written testimonial or comments provided by the user

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User Id	Int		Primary key	User Master	Unique identifier for each car rental
Fullname	Varchar	120	Not Null	User Master	The full name of the person renting the car
Dob	Varchar	100	Not Null	User Master	The user's birth date for age verification
City	Varchar	100	Not Null	User Master	The city where the person resides or where the car rental is being registered
Country	Varchar	100	Not Null	User Master	The country where the person resides or where the car rental is being registered
Vehicle Id	Int		Primary key	Vehicle Master	A unique identifier for each vehicle in the system
Vehicle Name	Varchar	150	Not Null	Vehicle Master	The title or name associated with the vehicle
Vehicles Brand	Int		Not Null	Vehicle Master	The brand or make of the vehicle (e.g., Toyota, Honda)
Vehicles Overview	Long text		Not Null	Vehicle Master	Unique identifier for each vehicle overview
Price per Day	Int		Not Null	Vehicle Master	Unique identifier for each pricing record
Fuel Type	Varchar	100	Not Null	Vehicle Master	The type of fuel the vehicle uses (e.g., Gasoline, Diesel)
Model Year	Int		Not Null	Vehicle Master	The manufacturing year of the vehicle
Seating Capacity	Int		Not Null	Vehicle Master	The maximum seating capacity of the vehicle
Vimage 1	Varchar	120	Not Null	Vehicle Master	It is view image of car
Vimage 2	Varchar	120	Not Null	Vehicle Master	It is view image of car
Vimage 3	Varchar	120	Not Null	Vehicle Master	It is view image of car
Vimage 4	Varchar	120	Not Null	Vehicle Master	It is view image of car
Vimage 5	Varchar	120	Not Null	Vehicle Master	It is view image of car
Air Conditioner	Int		Not Null	Vehicle Master	Indicates whether the vehicle has air conditioning
Power DoorLocks	Int		Not Null	Vehicle Master	Indicates whether the vehicle has power door locks
AntiLockBreakSystem	Int		Not Null	Vehicle Master	Indicates whether the vehicle is equipped with an anti-lock brake system
Break Assist	Int		Not Null	Vehicle Master	Indicates whether the vehicle has brake assist features
Power Steering	Int		Not Null	Vehicle Master	Indicates whether the vehicle has power steering
Driver AirBag	Int		Not Null	Vehicle Master	Indicates whether the vehicle is equipped with a driver airbag
Passenger AirBag	Int		Not Null	Vehicle Master	Indicates whether the vehicle is equipped with a passenger airbag
PowerWindows	Int		Not Null	Vehicle Master	Indicates whether the vehicle has power windows
CD player	Int		Not Null	Vehicle Master	Indicates whether the vehicle is equipped with a CD player
CentralLocking	Int		Not Null	Vehicle Master	Indicates whether the vehicle has central locking
CrashSensor	Int		Not Null	Vehicle Master	Indicates whether the vehicle is equipped with a crash sensor
LeatherSeats	Int		Not Null	Vehicle Master	Indicates whether the vehicle has leather seats
Reg Date	Timestamp		Not Null	User Master, Vehicle Master	The registration date of the vehicle
Payment_Id	Int		Primary Key	Payment Master	Unique identifier for each payment transaction
Order_Id	Int		Not Null	Payment Master	Unique identifier for the rental order associated with the payment
Name On Card	Varchar		Not Null	Payment Master	The name printed on the credit/debit card used for payment
Card Number	Int		Not Null	Payment Master	Masked or encrypted credit/debit card number
Expiry Date	Timestamp		Not Null	Payment Master	Expiry date of the credit/debit card
CVV	Int		Not Null	Payment Master	Card Verification Value for security purposes
Amount	Int		Not Null	Payment Master	The total amount paid by the customer for the rental order
Payment Date	Timestamp		Not Null	Payment Master	Date and time when the payment was made

6.2 Database Design/Table Design

Table: Admin

Description: This table provide the information about admin

Field name	Data type	Size	Constraint
Id	Int	-	Primary key
User name	Text	100	Not Null
Password	Text	100	Not Null
Updating date	Date	-	Not Null

Table: Booking

Description: This table provide the information about booking

Field name	Data type	Size	Constraint
Id	Int	-	Primary key
Booking number	Int	-	Not Null
User email	Text	100	Not Null
Vehicle Id	Int	-	Not Null
From Date	Date	20	Not Null
To Date	Date	20	Not Null
Message	Text	255	Not Null
Status	Int	-	Not Null
Posting date	Date	-	Not Null
Last Updation Date	Date	-	Not Null

Table: Brand

Description: This table provide the information about brands

Field name	Data type	Size	Constraint
Id	Int	-	Primary key
Brand name	Text	120	Not Null
Creation date	Date	-	Not Null
Updation date	Date	-	Not Null

Table: ContactUsInfo

Description: This table provide the information about ContactUsInfo

Field name	Data type	Size	Constraint
Id	Int	-	Primary key
Address	Text	120	Not Null
Email Id	Text	255	Not Null
Contac No	Int	-	Not Null

Table: ContactUsQuery

Description: This table provide the information about ContactUsQuery

Field name	Type	Size	Constraint
Id	Int	-	Primary key
Name	Text	120	Not Null
Email Id	Text	255	Not Null
Contact Number	Int	11	Not Null
Message	Text	-	Not Null
Posting Date	Date	-	Not Null
Status	Int	-	Not Null

Table: Pages

Description: This table provide the information about pages

Field name	Data type	Size	Constraint
Id	Int	-	Primary key
Page Name	Text	255	Not Null
Type	Text	255	Not Null
Detail	Text	-	Not Null

Table: Subscription

Description: This table provide the information about subscription

Field name	Data type	Size	Constraint
Id	Int	-	Primary key
Subscription Email	Text	120	Not Null
Posting Date	Date	-	Not Null

Table: Testimonial

Description: This table provide the information about testimonial

Field name	Data type	Size	Constraint
Id	Int	-	Primary key
User email	Text	100	Not Null
Testimonial	Text	-	Not Null
Updation_Date	Date	-	Not Null
Status	Int	-	Not Null

Table: User

Description: This table provide the information about user

Field name	Type	Size	Constraint
Id	Int	-	Primary key
Full Name	Text	120	Not Null
Email_Id	Text	255	Foreign Key
Password	Text	11	Not Null
Contact No	Int	11	Not Null
Dob	Date	100	Not Null
Address	Text	255	Not Null
City	Text	100	Not Null
Country	Text	100	Not Null
Regdate	Date	-	Not Null
Updation_Date	Date	-	Not Null

Table: Vehicles

Description: This table is provide the information about vehicles

Field name	Data type	Size	Constraint
Id	Int	-	Primary key
Vehicle Title	Text	150	Not Null
Vehicles Brand	Int	-	Not Null
Vehicles Overview	Text	-	Not Null
PriceperDay	Int	-	Not Null
Fuel Type	Text	100	Not Null
Model Year	Int	-	Not Null
Seating Capacity	Int	-	Not Null
Vimage 1	Text	120	Not Null
Vimage 2	Text	120	Not Null
Vimage 3	Text	120	Not Null
Vimage 4	Text	120	Not Null
Vimage 5	Text	120	Not Null
Air Conditioner	Int	-	Not Null
PowerDoorLocks	Int	-	Not Null
AntiLockBreakSystem	Int	-	Not Null
Brake Assist	Int	-	Not Null
Power Steering	Int	-	Not Null
Driver Airbag	Int	-	Not Null
Passenger Airbag	Int	-	Not Null
Power Windows	Int	-	Not Null
CD player	Int	-	Not Null
Central Locking	Int	-	Not Null
Crash Sensor	Int	-	Not Null
Leather Seats	Int	-	Not Null
Regdate	Date	-	Not Null
Updation Date	Date	-	Not Null

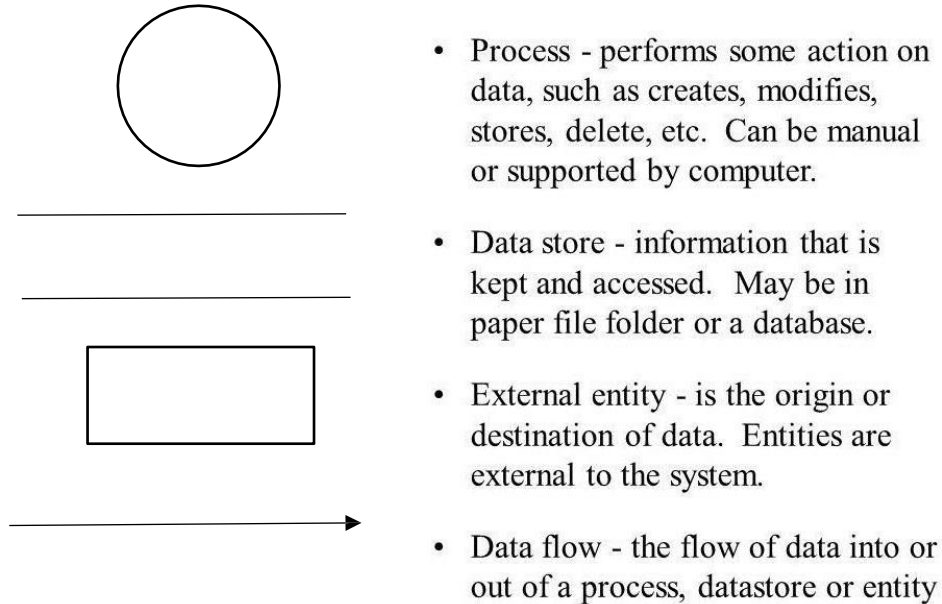
Table: Payment

Description: This table provide the information about payment

Field name	Data type	Size	Constraint
Id	Int	-	Primary key
Order_Id	Int	-	Not Null
Name_On_Card	Text	255	Not Null
Card_Number	Int	16	Not Null
Expiry_Date	Int	5	Not Null
CVV	Int	4	Not Null
Amount	Int	10	Not Null
Payment Date	Date	-	Not Null

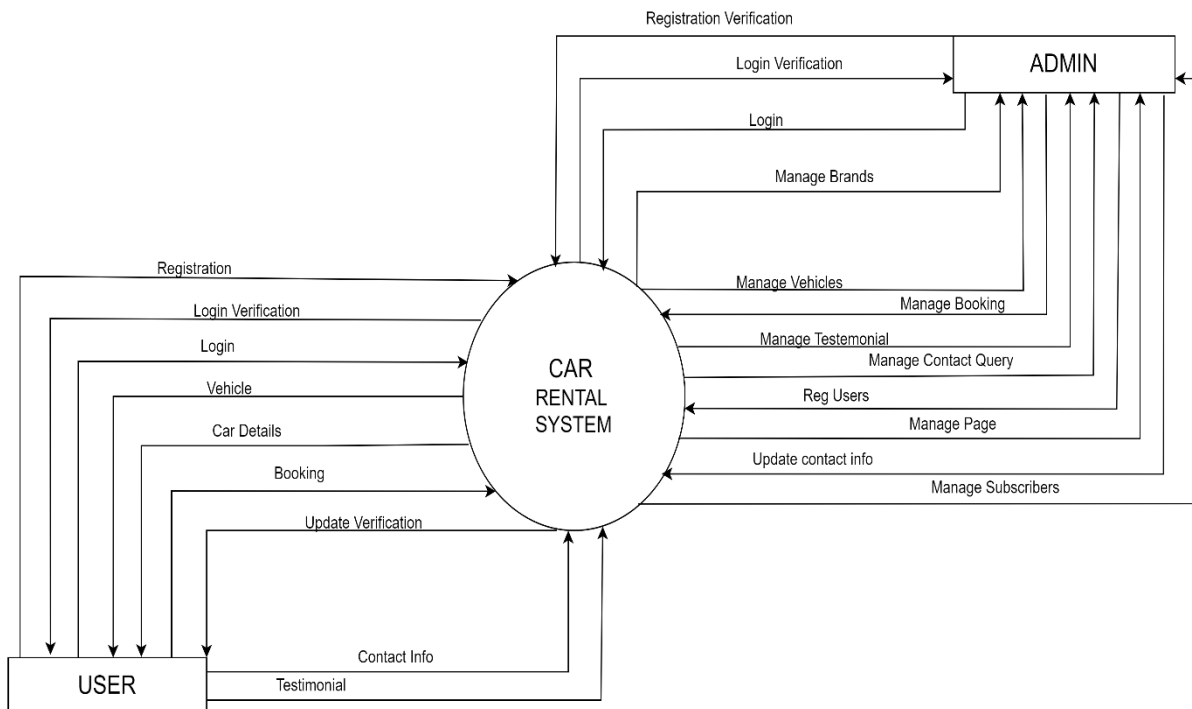
6.3 Data Flow Diagram

Data Flow Diagram Symbols

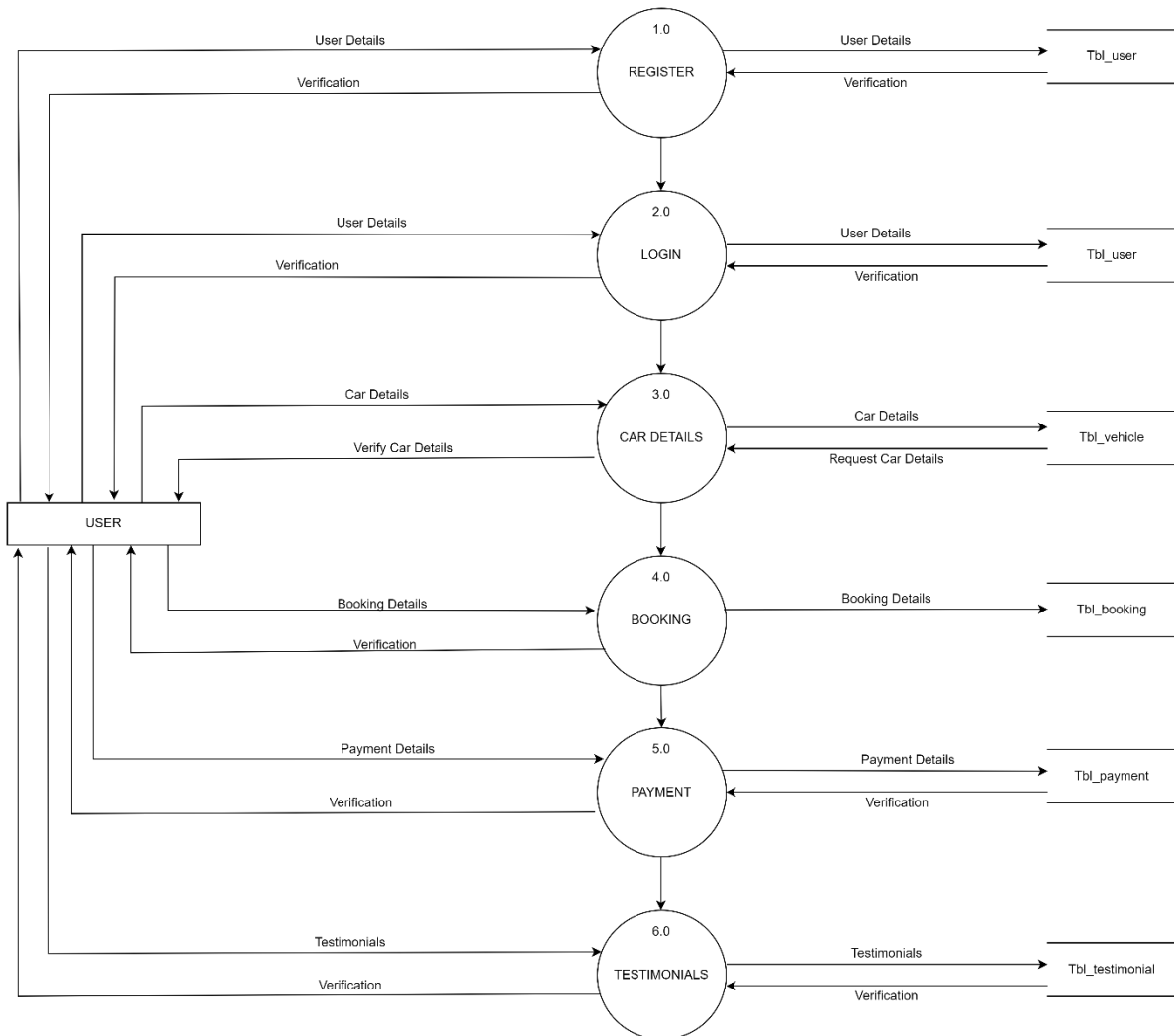


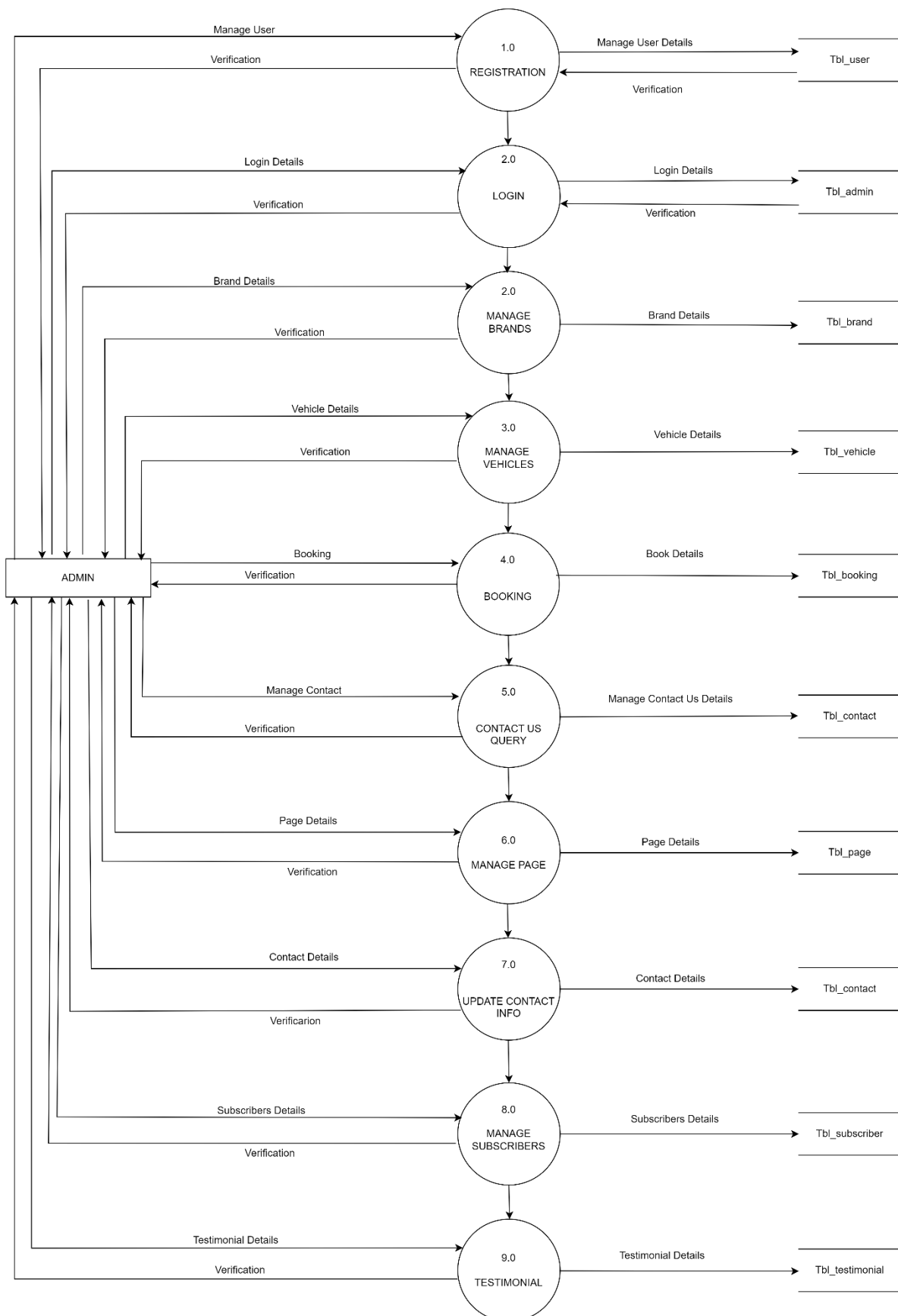
Several

CONTEXT LEVEL DFD

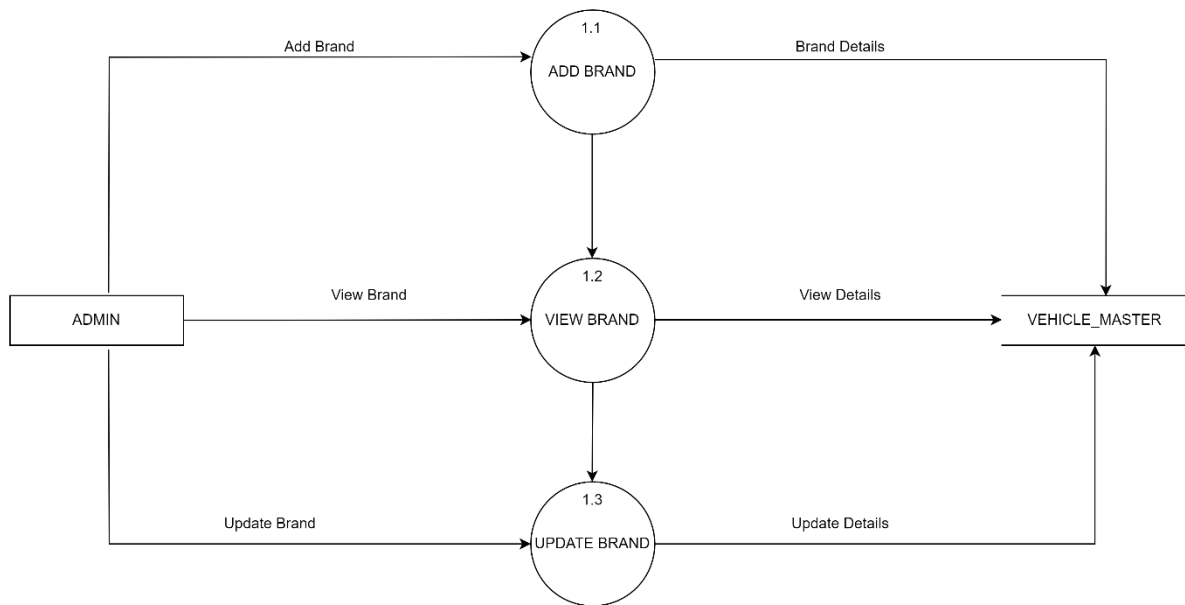


1ST LEVEL DFD OF USER MASTER

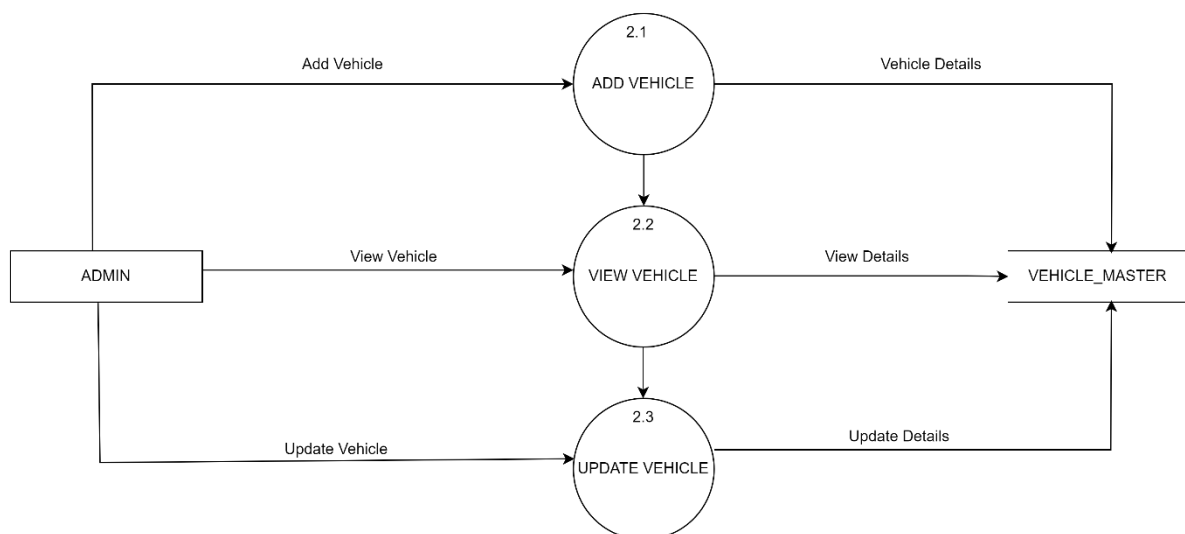


1st LEVEL DFD OF ADMIN MASTER

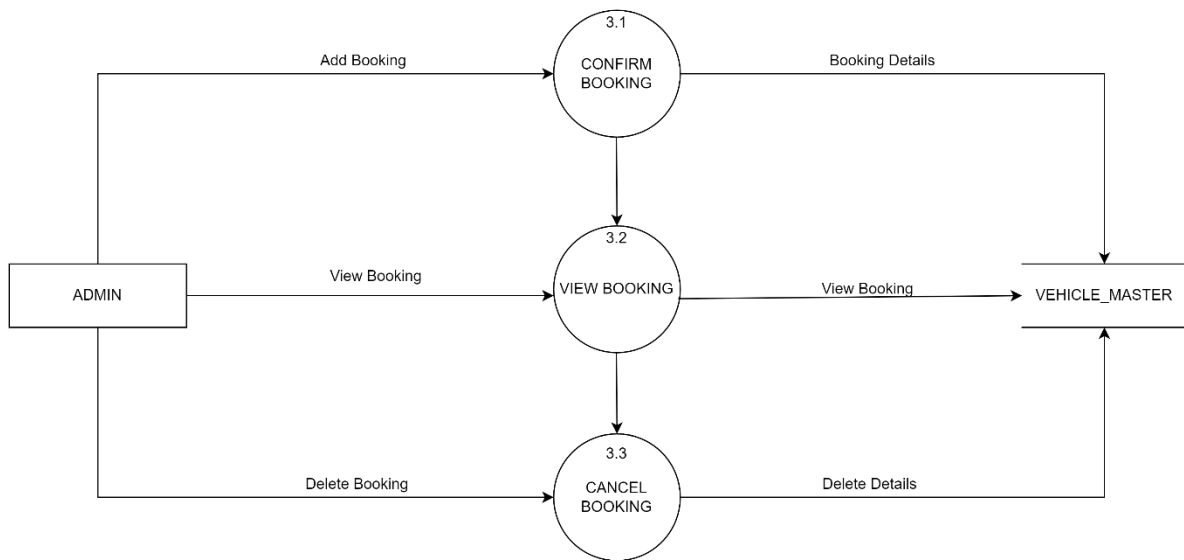
2nd LEVEL DFD OF ADMIN MASTER(BRAND)



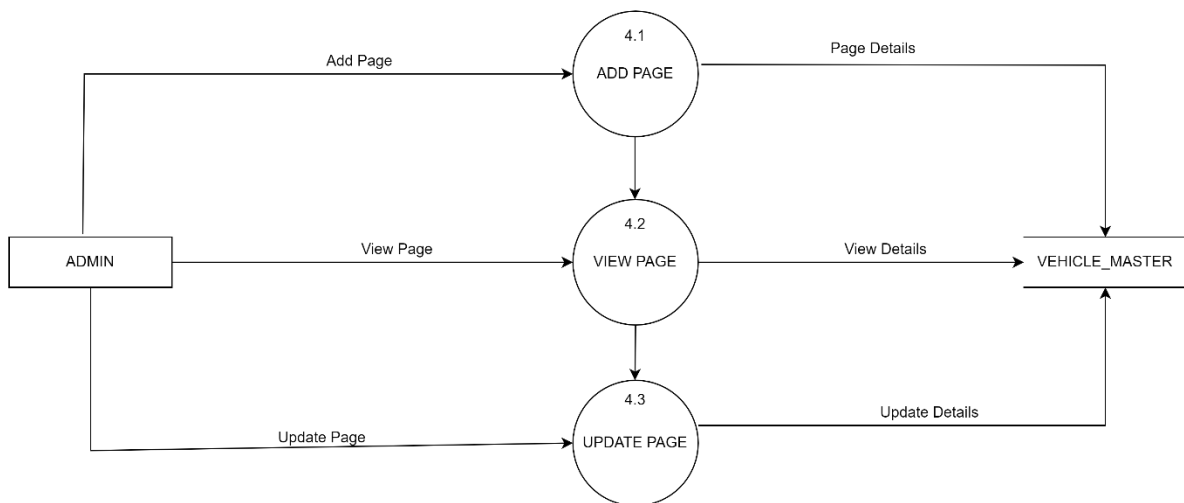
2nd LEVEL DFD OF ADMIN MASTER(VEHICLE)



2nd LEVEL DFD OF ADMIN MASTER(BOOKING)

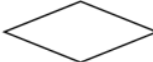


2nd LEVEL DFD OF ADMIN MASTER(PAGE)

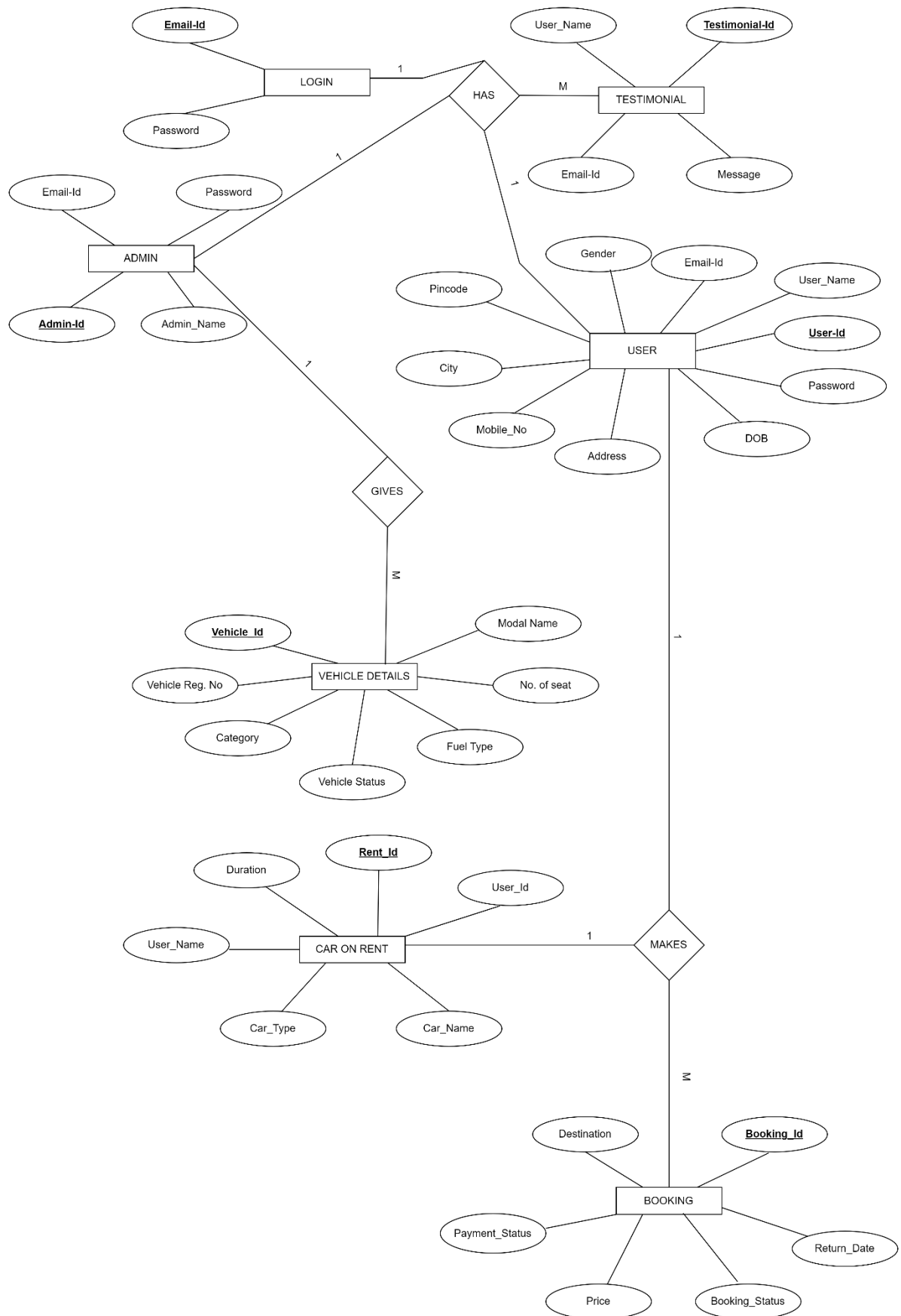


6.4 E-R Diagram

The set of symbols we have used in E-R diagrams. There is no universal standard for E-R diagram notation, and different books and E-R diagram software are different notations. Figure indicates some of the alternative notations that are widely used. An entity set may be represented as a box with the name outside, and the attributes.

Name	Symbol	Meaning
Oval		Shows different attributes
Rectangle		Shows entity set
Diamond		Show relationship among entity set
Line		Links entity set to attributes & entity set to relationship

CAR RENTAL MANAGEMENT SYSTEM

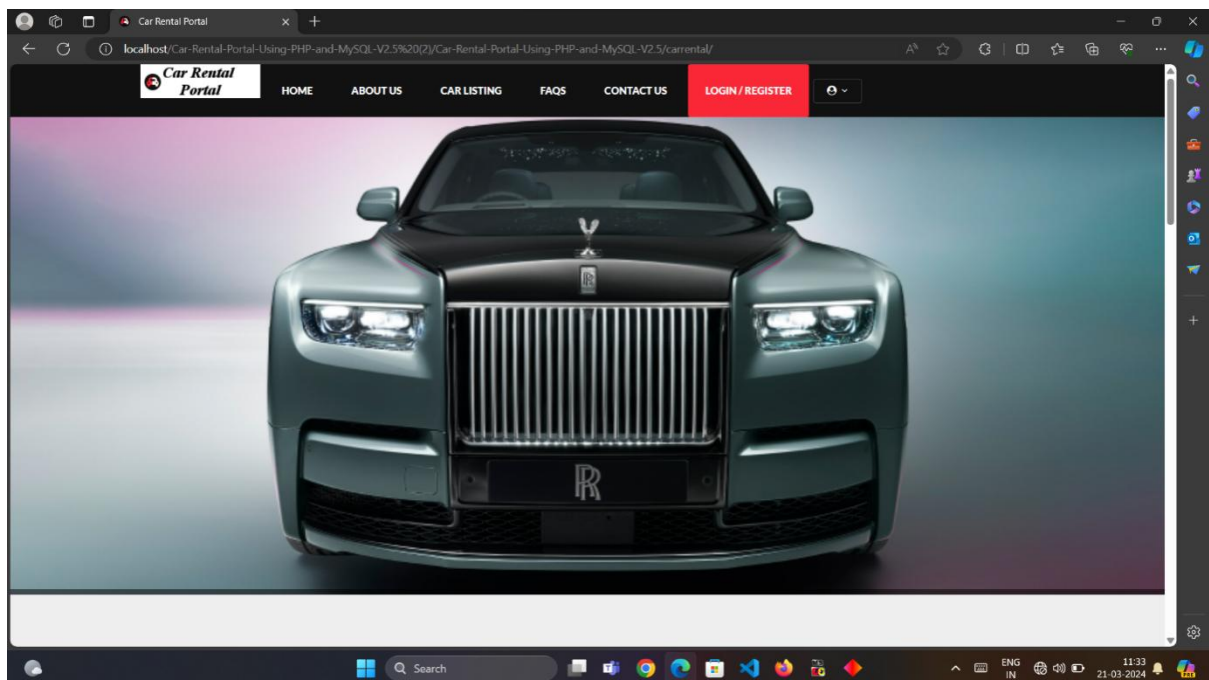


7. SYSTEM MODULES

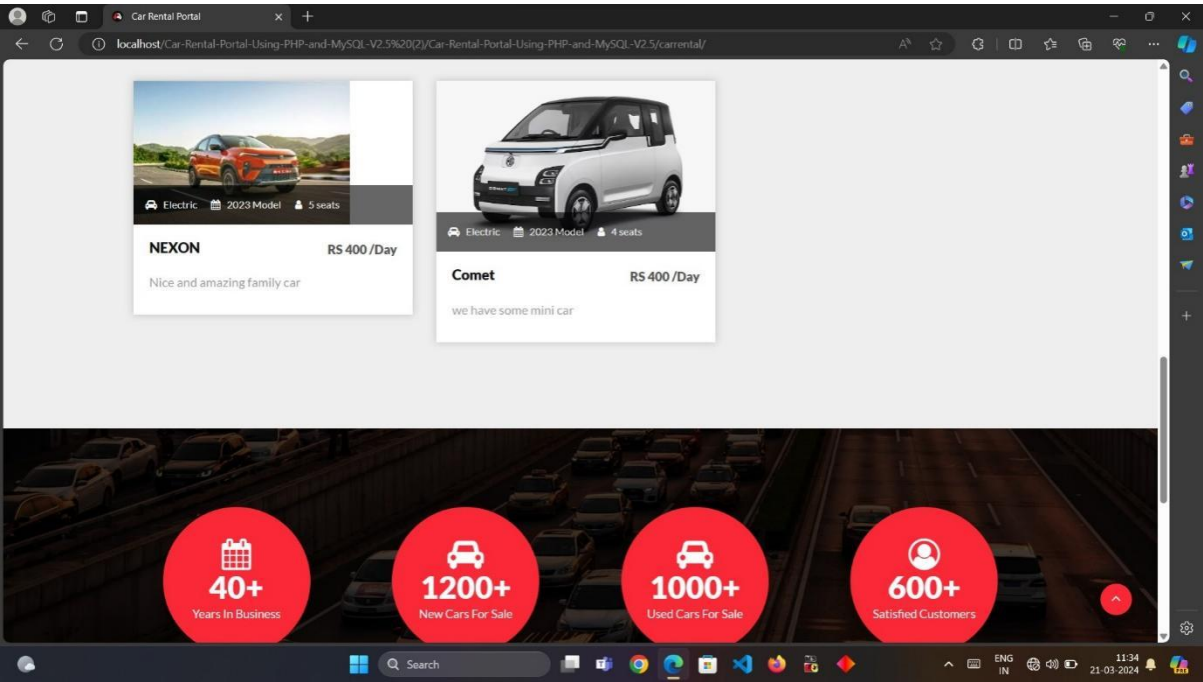
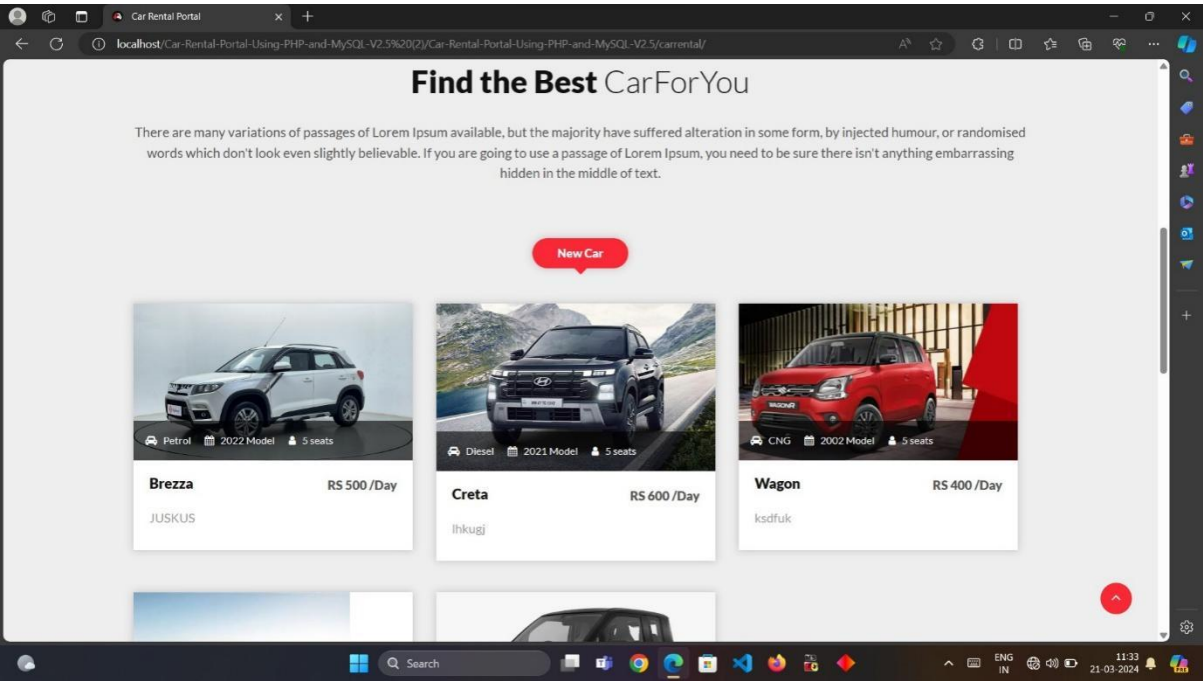
7.1 SCREEN LAYOUT

USER SIDE

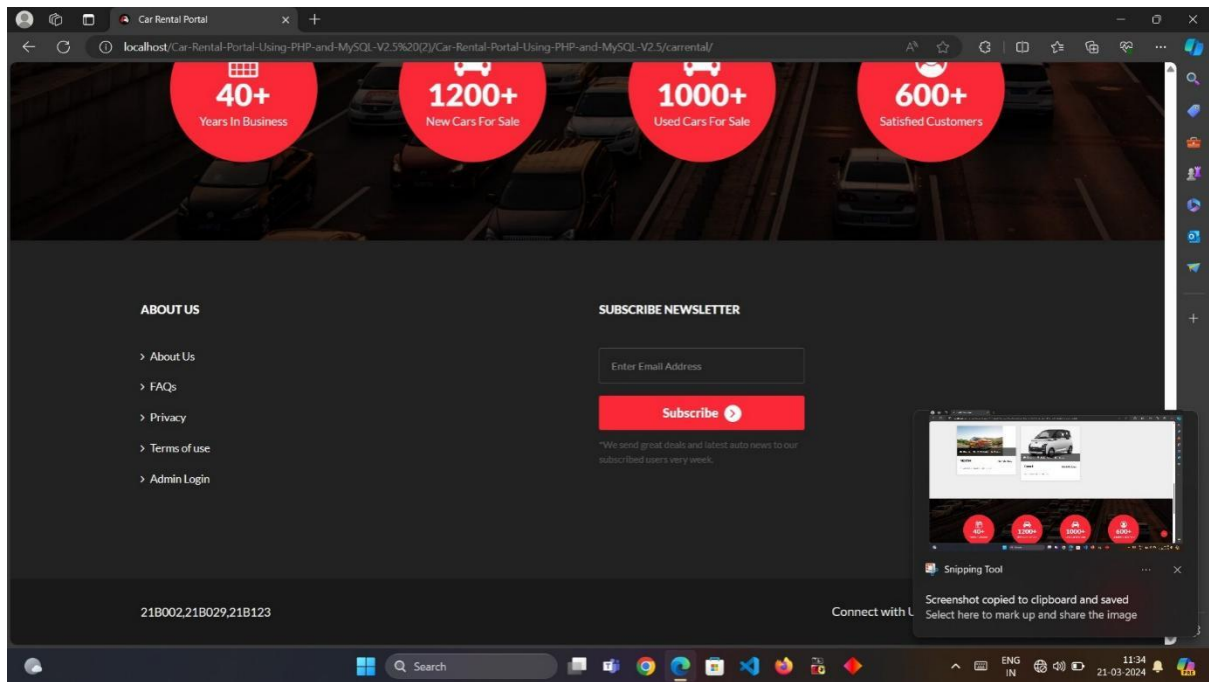
- User Home Page: Here user can see home page of website.



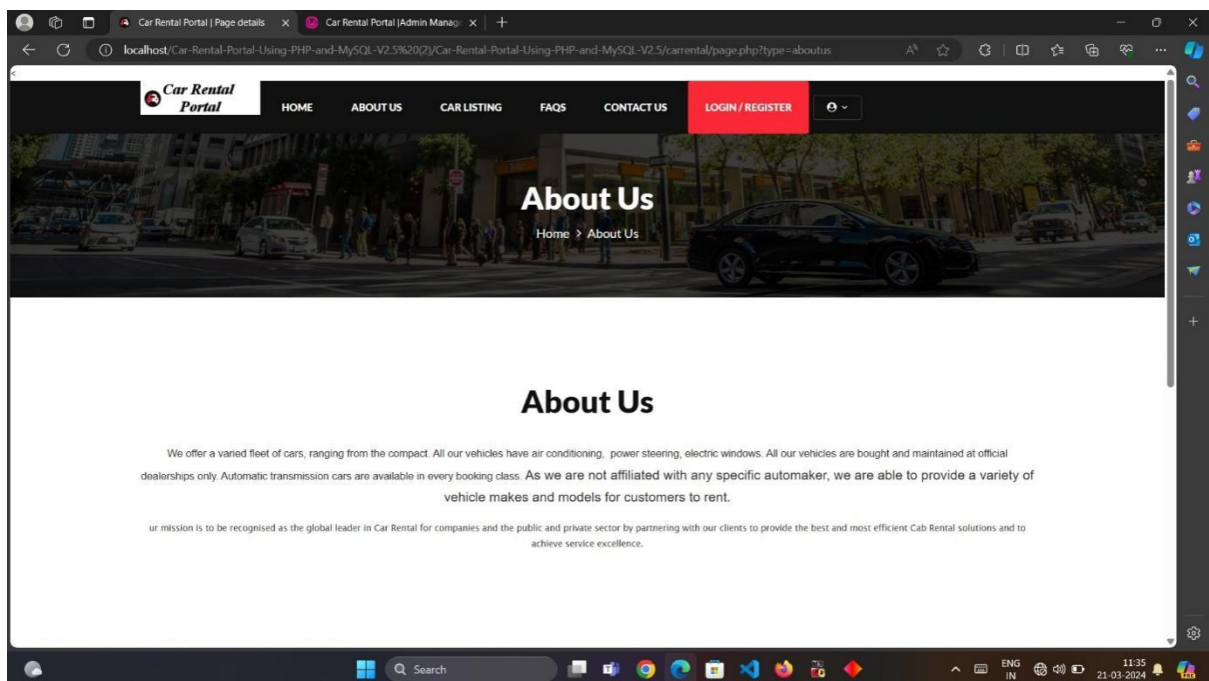
CAR RENTAL MANAGEMENT SYSTEM



CAR RENTAL MANAGEMENT SYSTEM

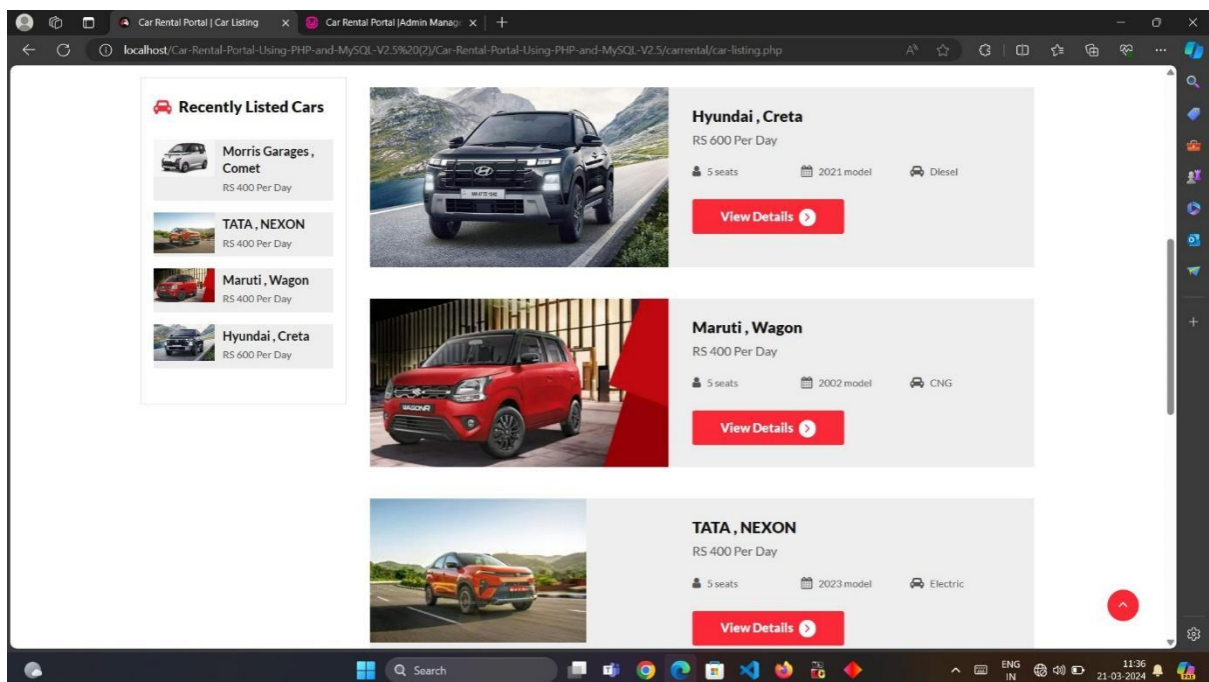
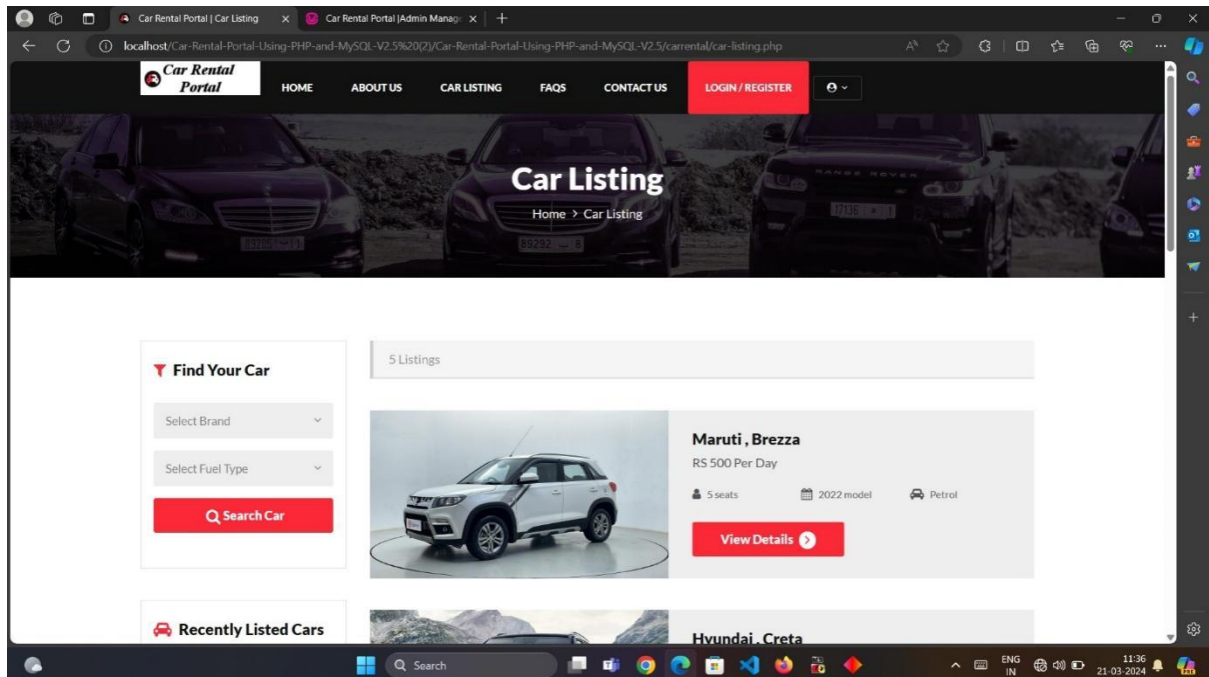


- User About Us Page: Here user can see about us page.

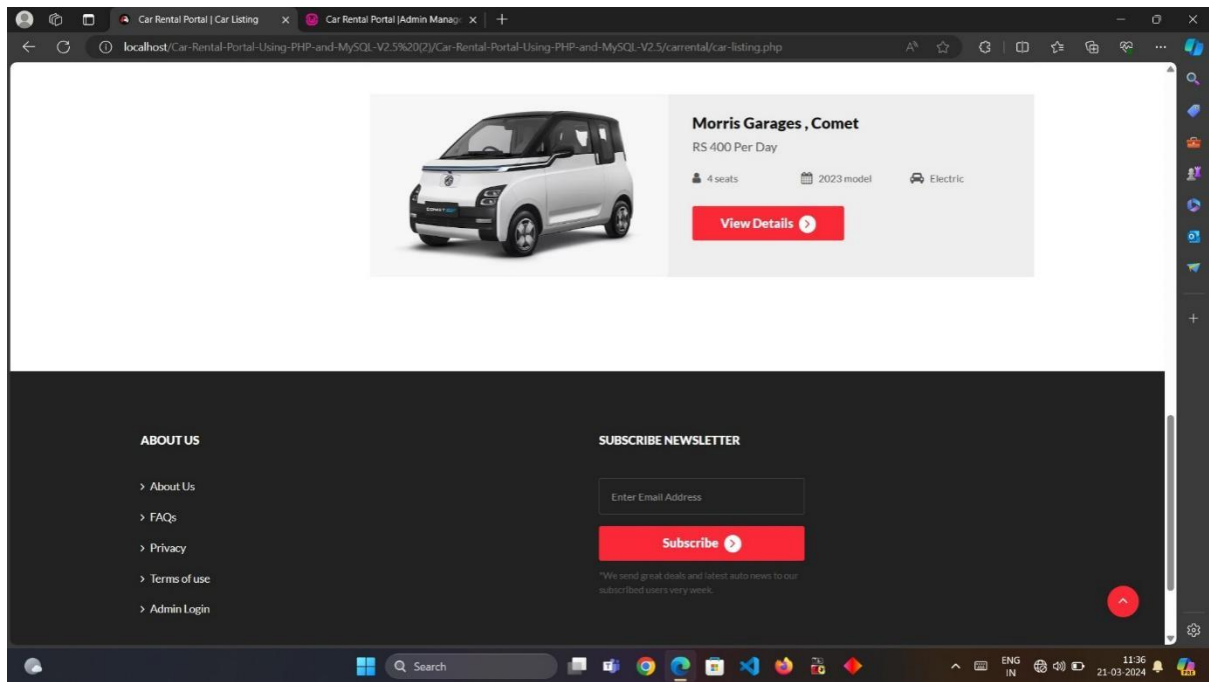


CAR RENTAL MANAGEMENT SYSTEM

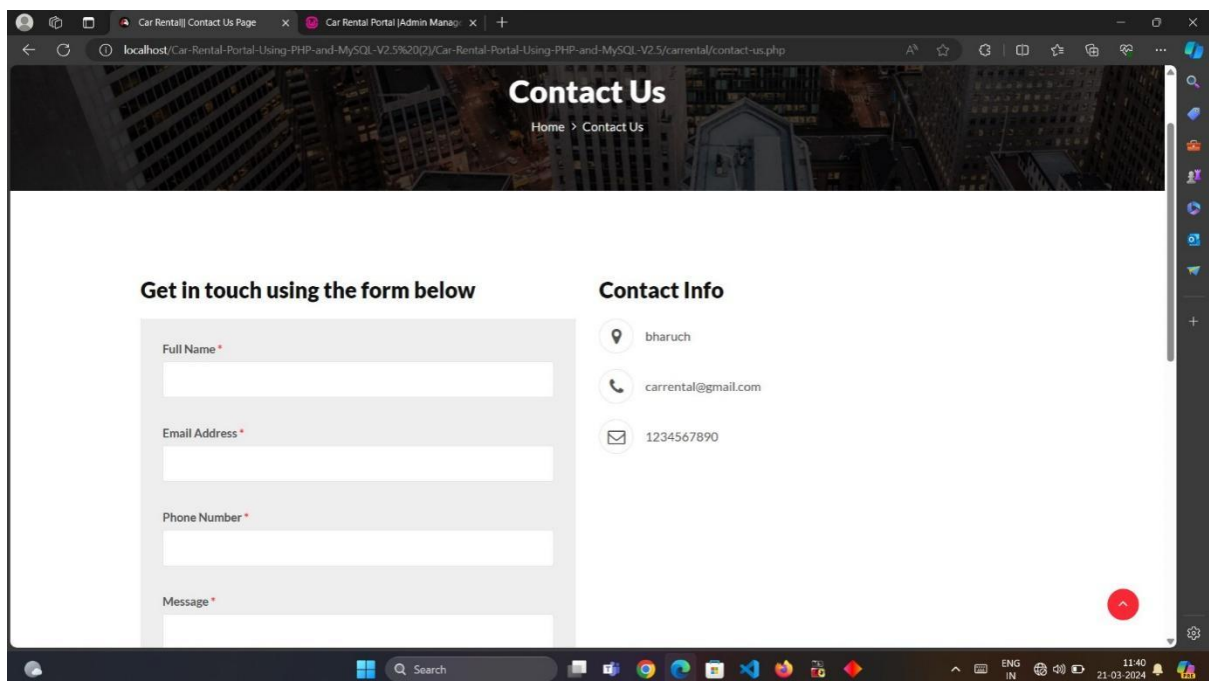
- User Car Listing Page: Here user can see listed cars with fuel, CNG + petrol, diesel.



CAR RENTAL MANAGEMENT SYSTEM



- **Contact Us Page:** Here user can contact us if any query come.



CAR RENTAL MANAGEMENT SYSTEM

Get in touch using the form below

Full Name *

Email Address *

Phone Number *

Message *

Send Message

Contact Info

bharuch

carrental@gmail.com

1234567890

- Sign up page: Here user can register the page.

Sign Up

dhyey

8780804891

dhyey@gmail.com

Email available for Registration.

☒ I Agree with Terms and Conditions

Sign Up

Already got an account? [Login Here](#)

Recently Listed Cars

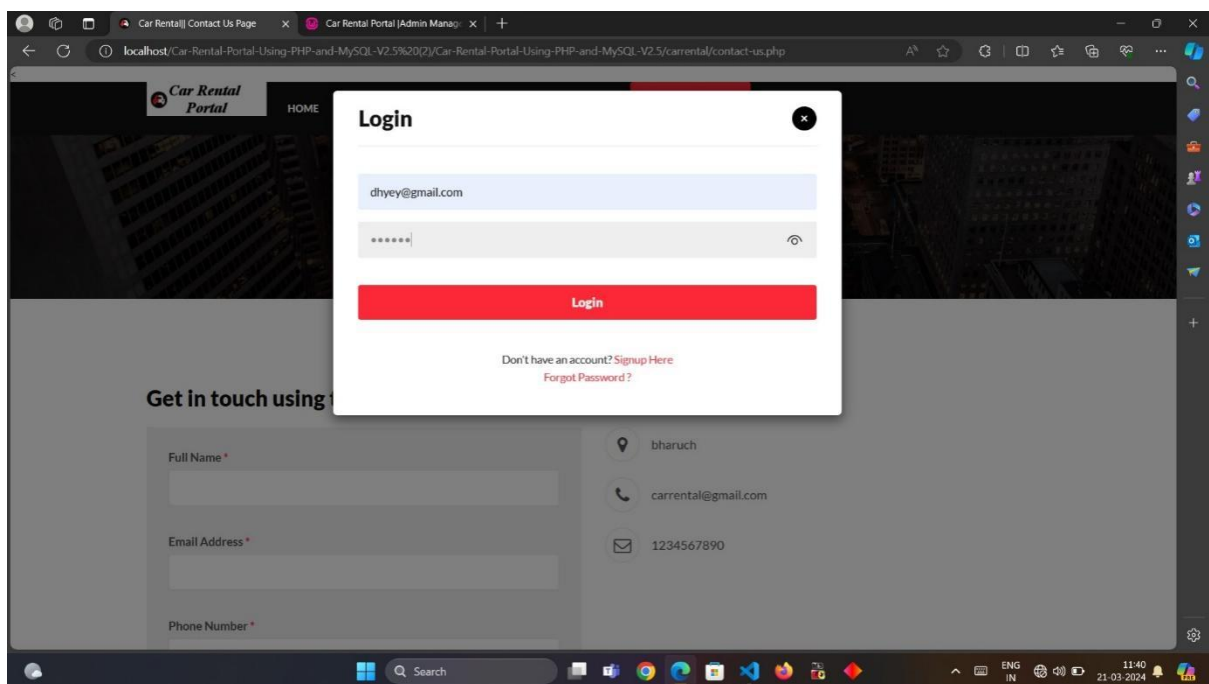
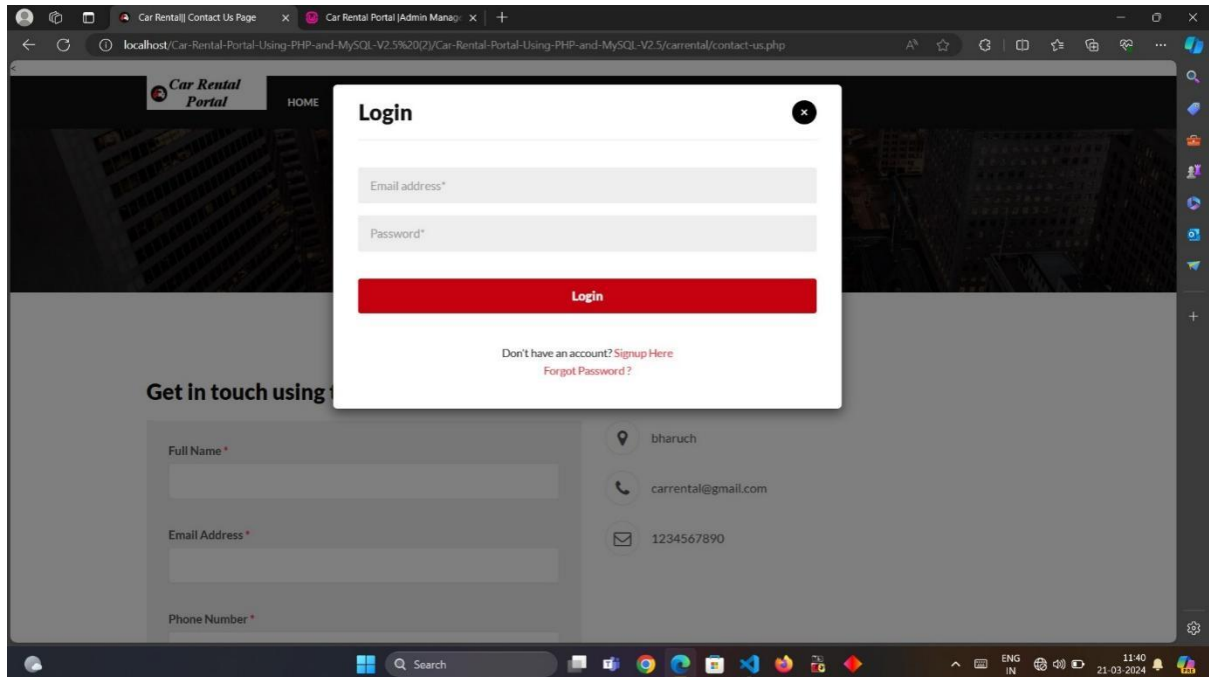
Maruti, Wagon
RS 400 Per Day

Hyundai, Creta
RS 600 Per Day

Maruti, Brezza
RS 500 Per Day

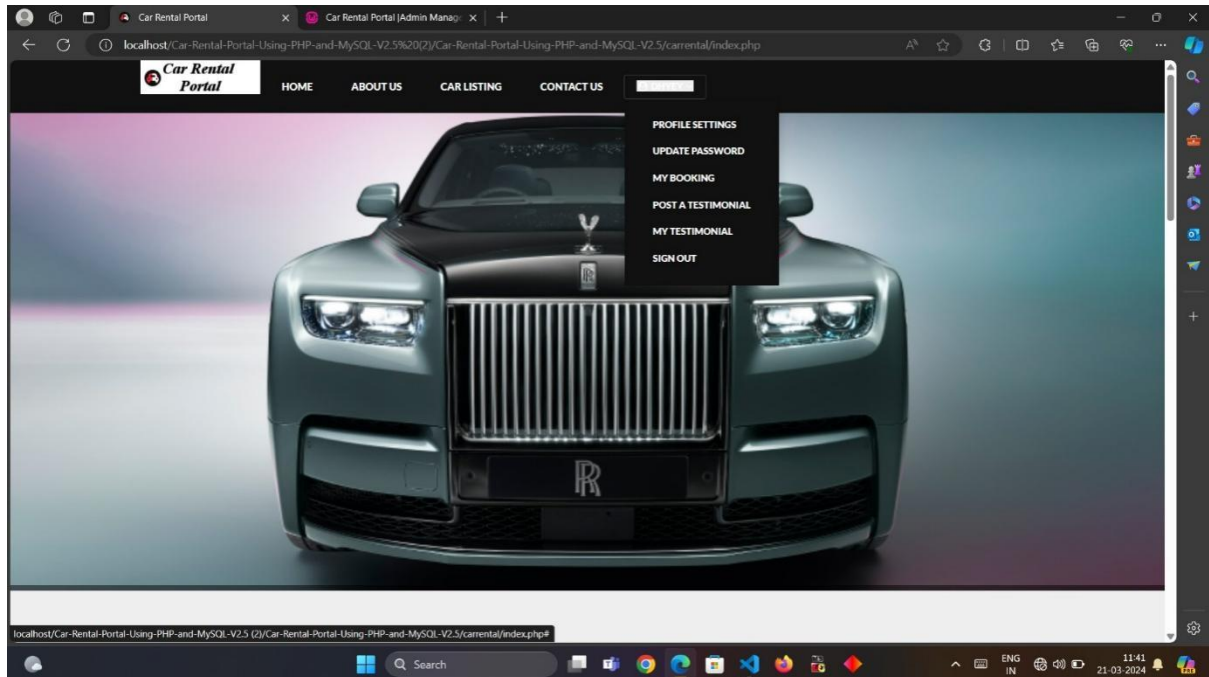
CAR RENTAL MANAGEMENT SYSTEM

- **User Login Page:** Here user can login if they get registered the page.

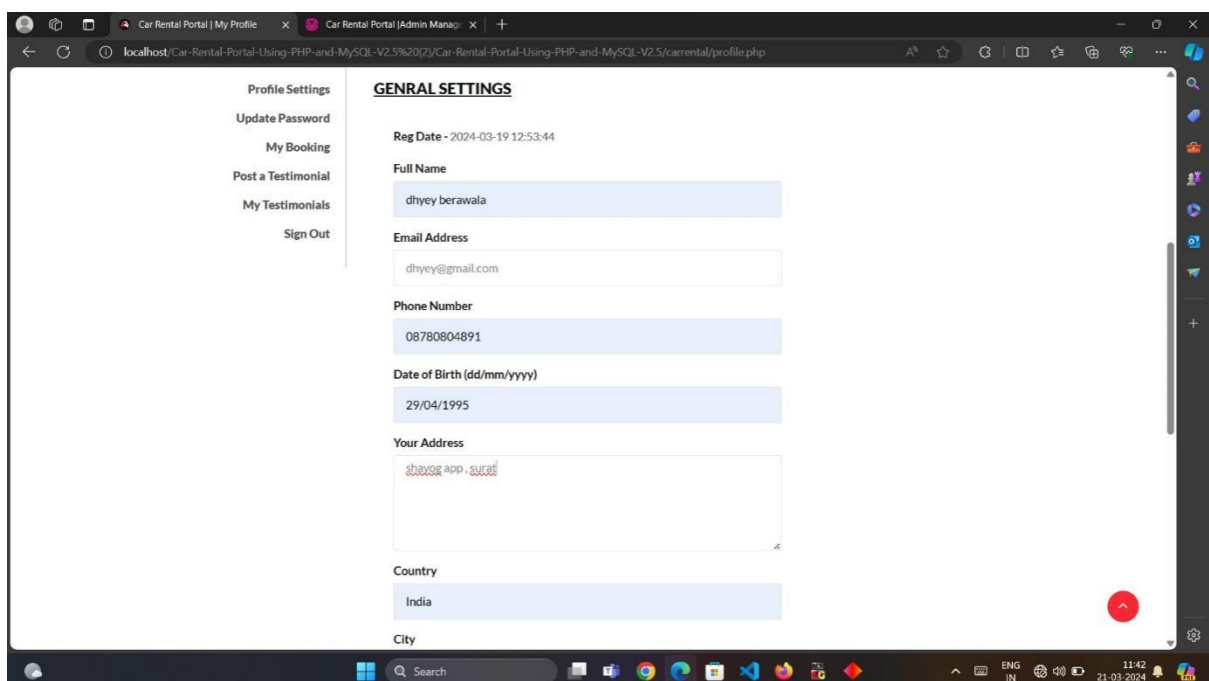


CAR RENTAL MANAGEMENT SYSTEM

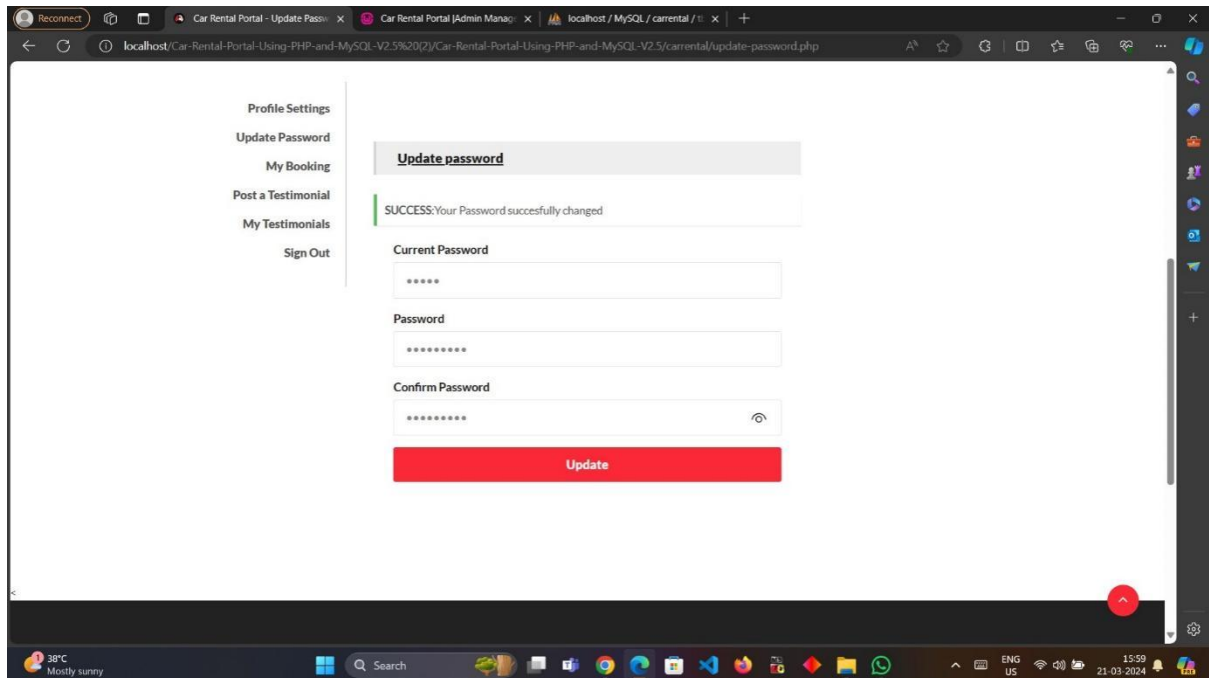
- User Profile Page: Here user can see details of their profile.



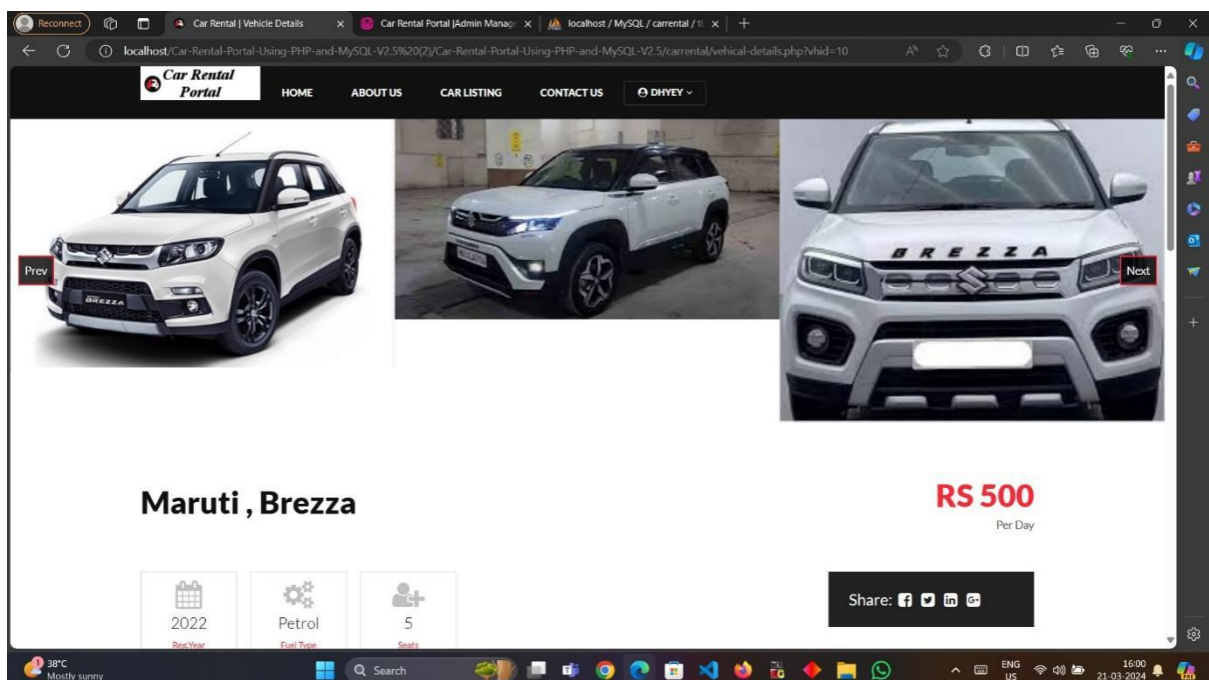
- User Profile Settings Page: Here User can see their profile.



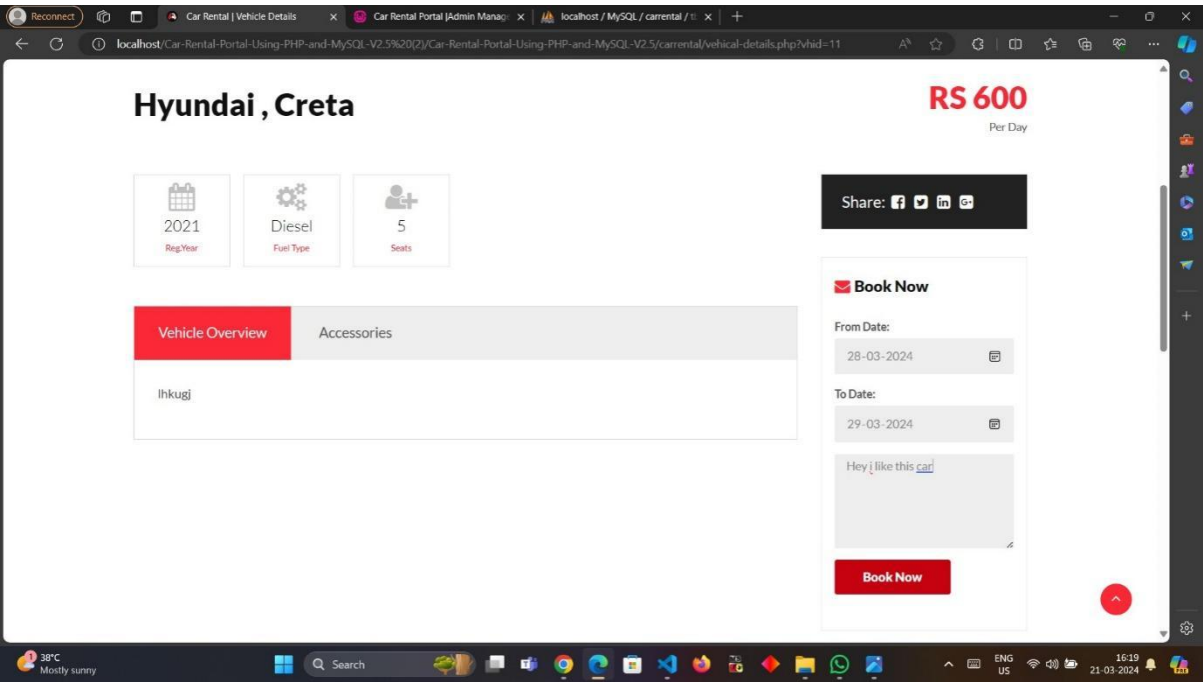
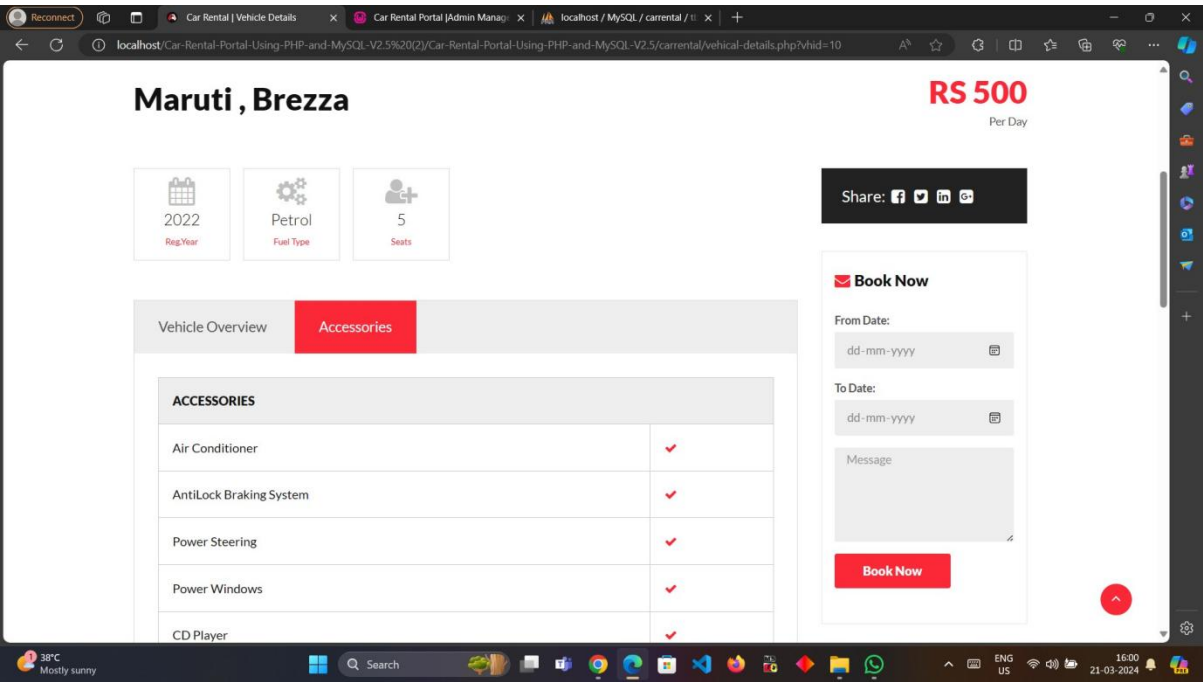
- Update Password Page: User can update their password.



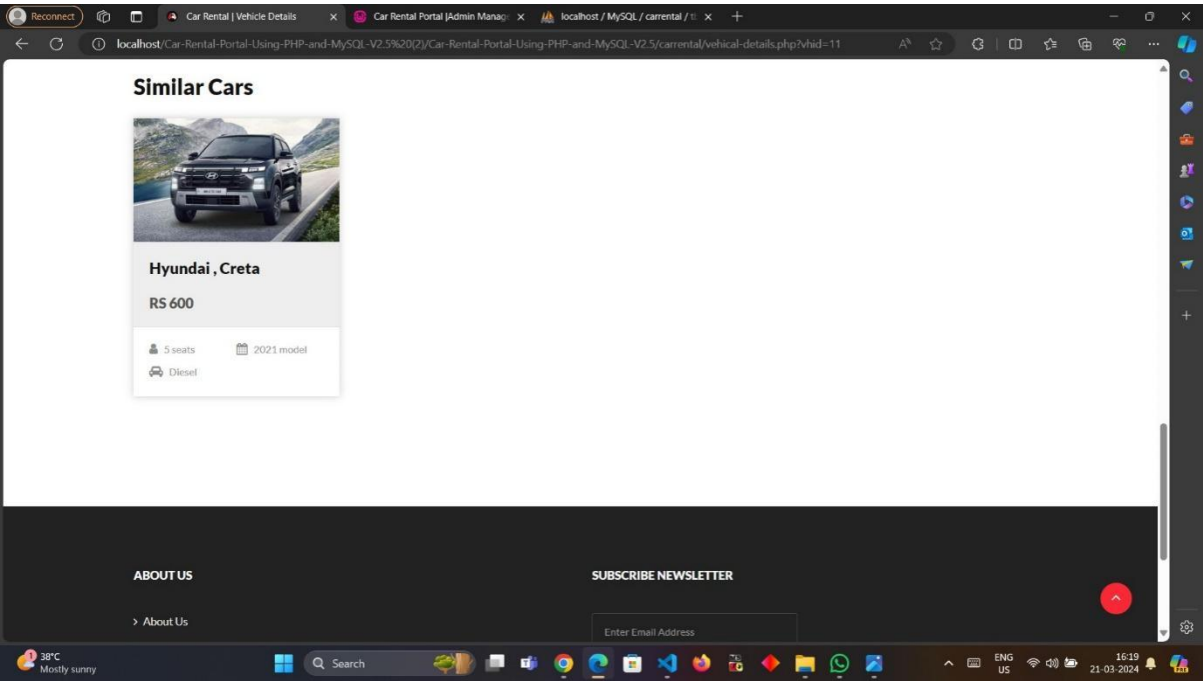
- Car Listing Profile Page: User can see car details.



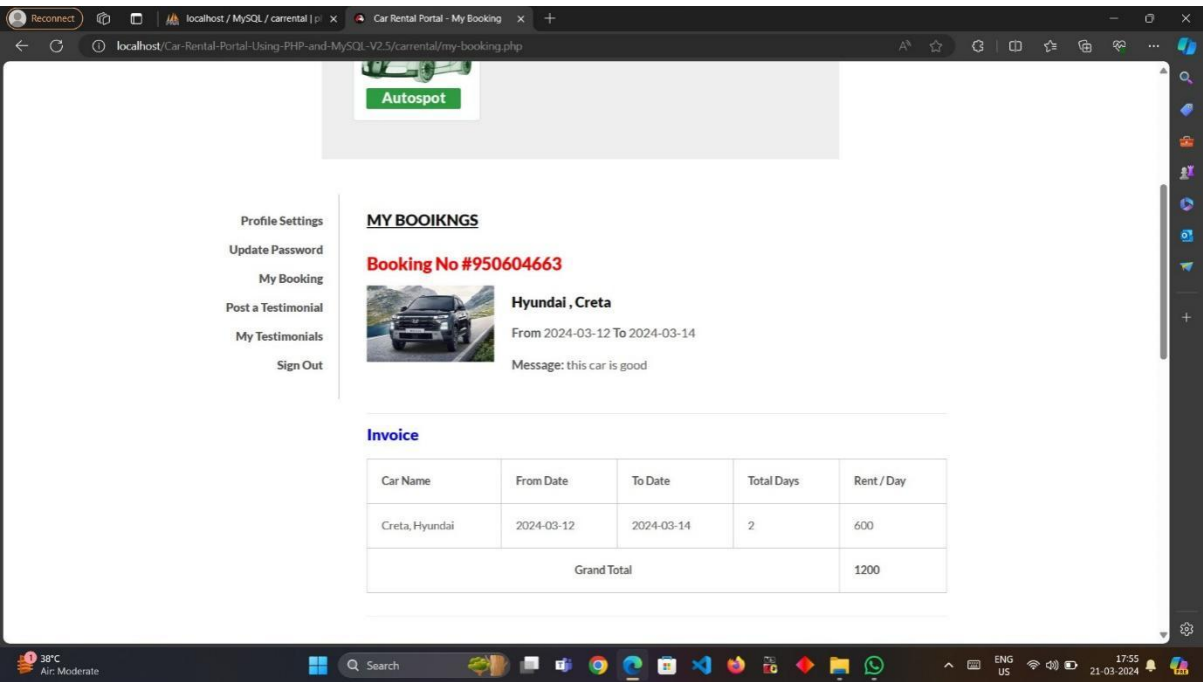
CAR RENTAL MANAGEMENT SYSTEM



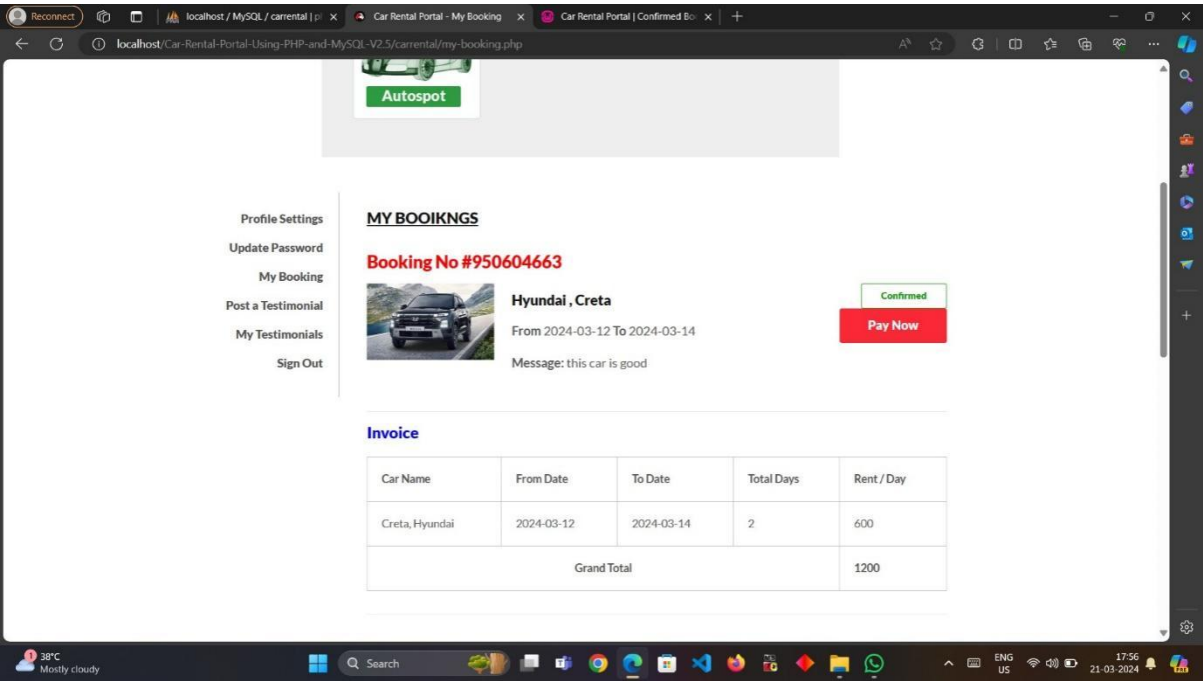
CAR RENTAL MANAGEMENT SYSTEM



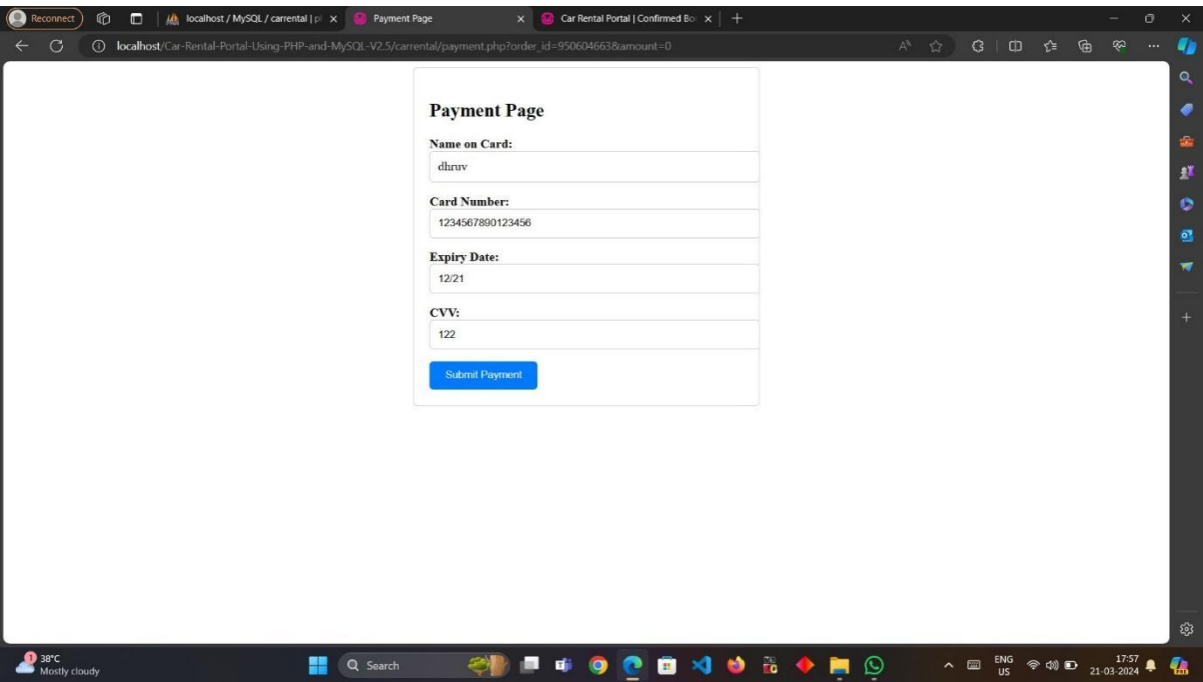
- User My Booking Page: User booking page.



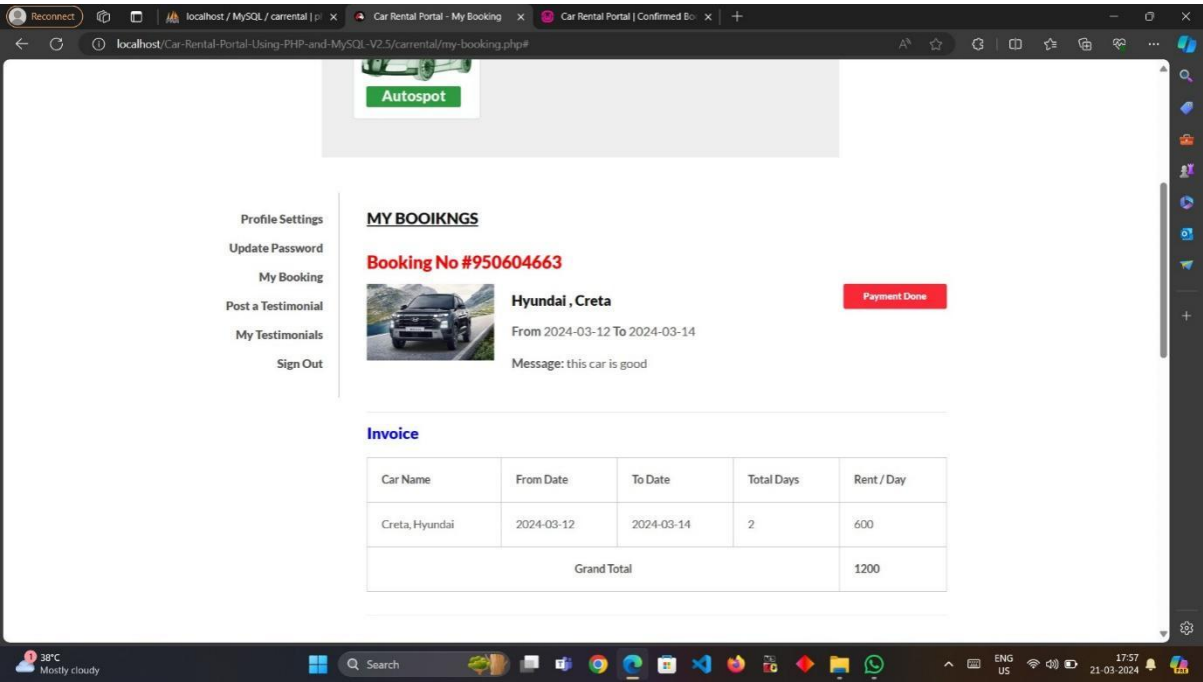
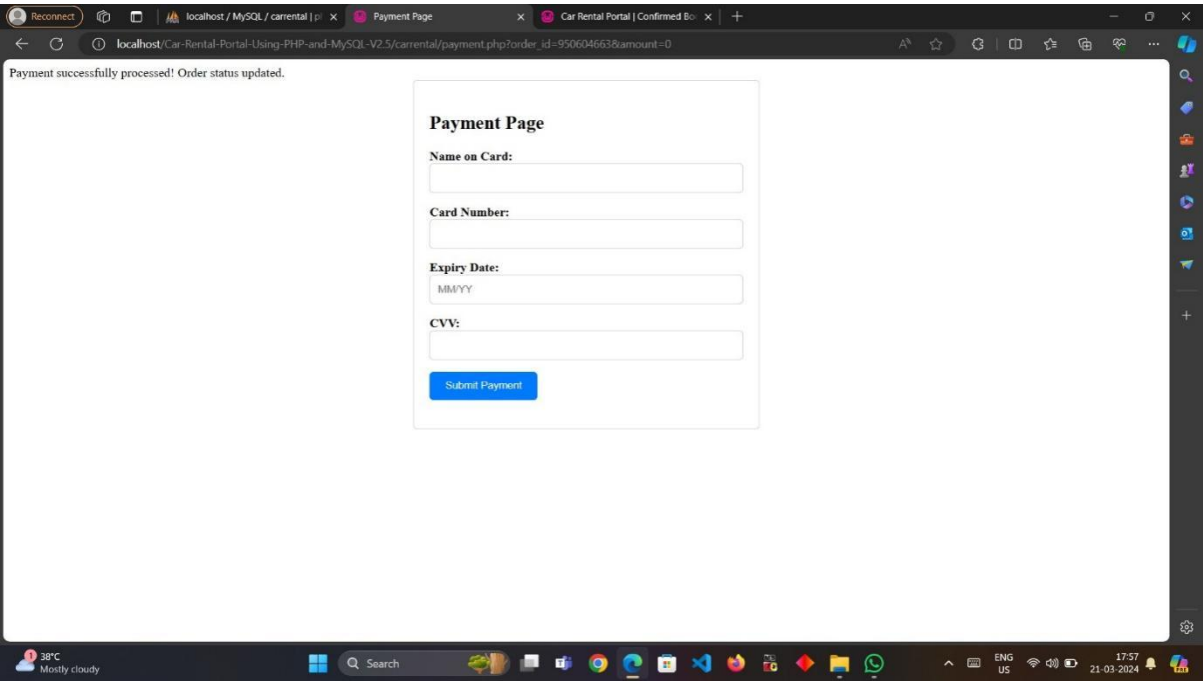
CAR RENTAL MANAGEMENT SYSTEM



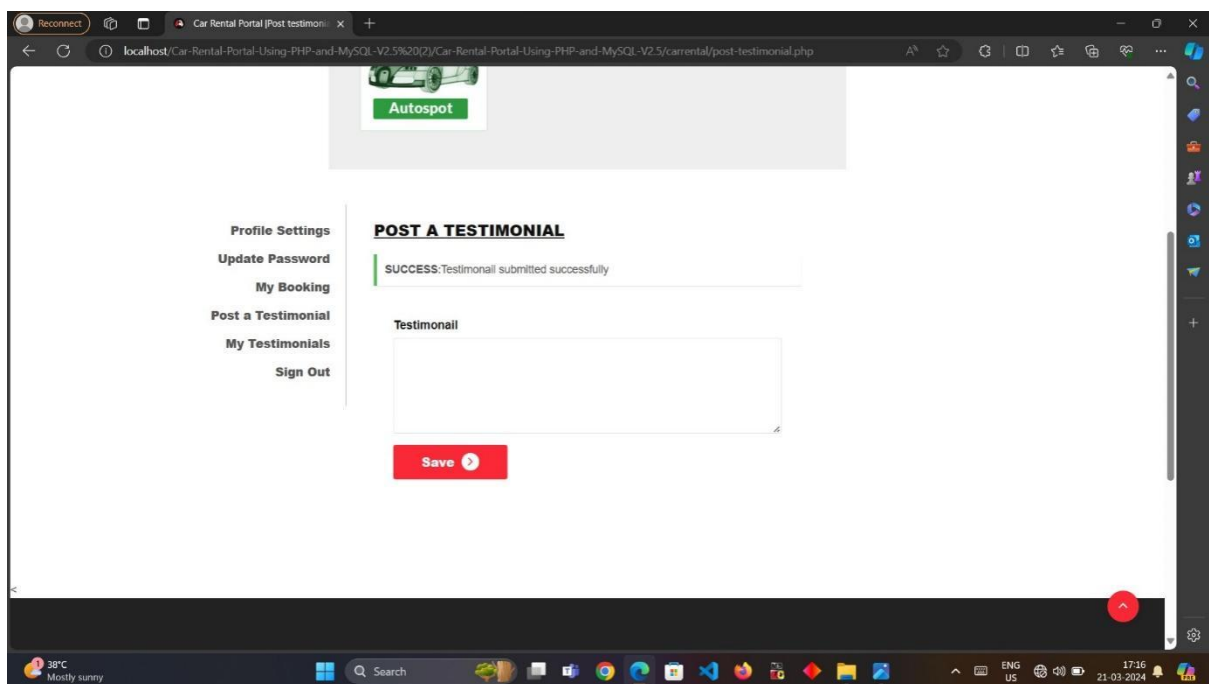
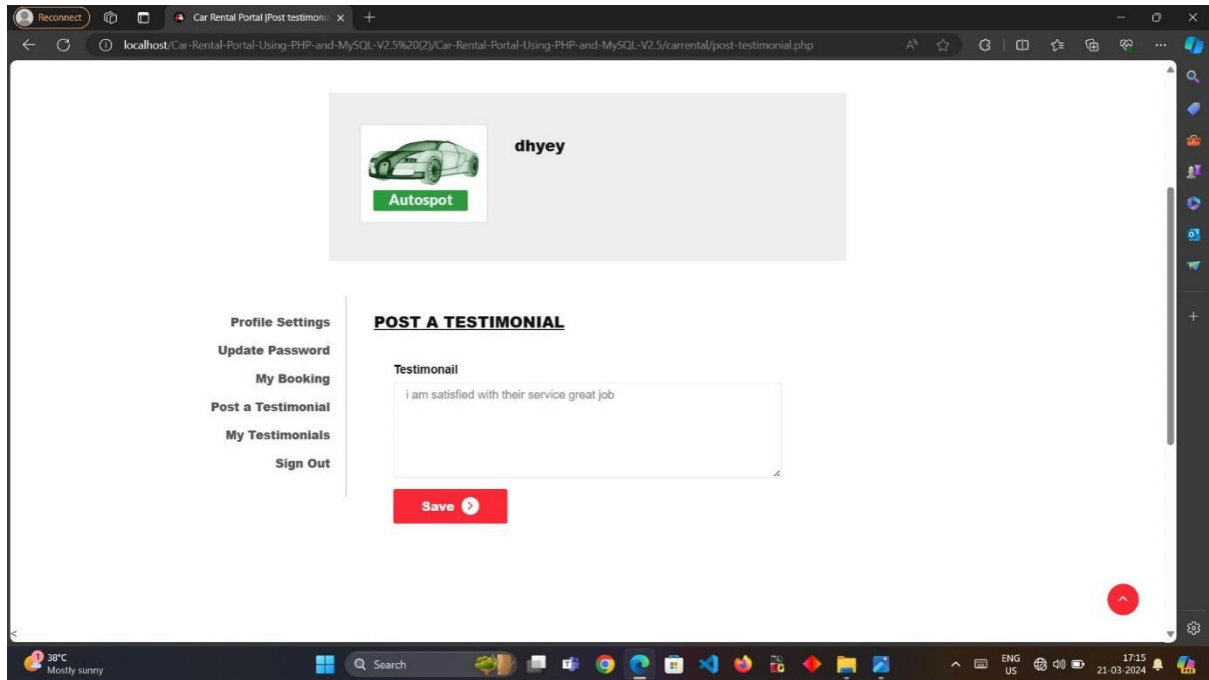
- Payment Page: User payment page.



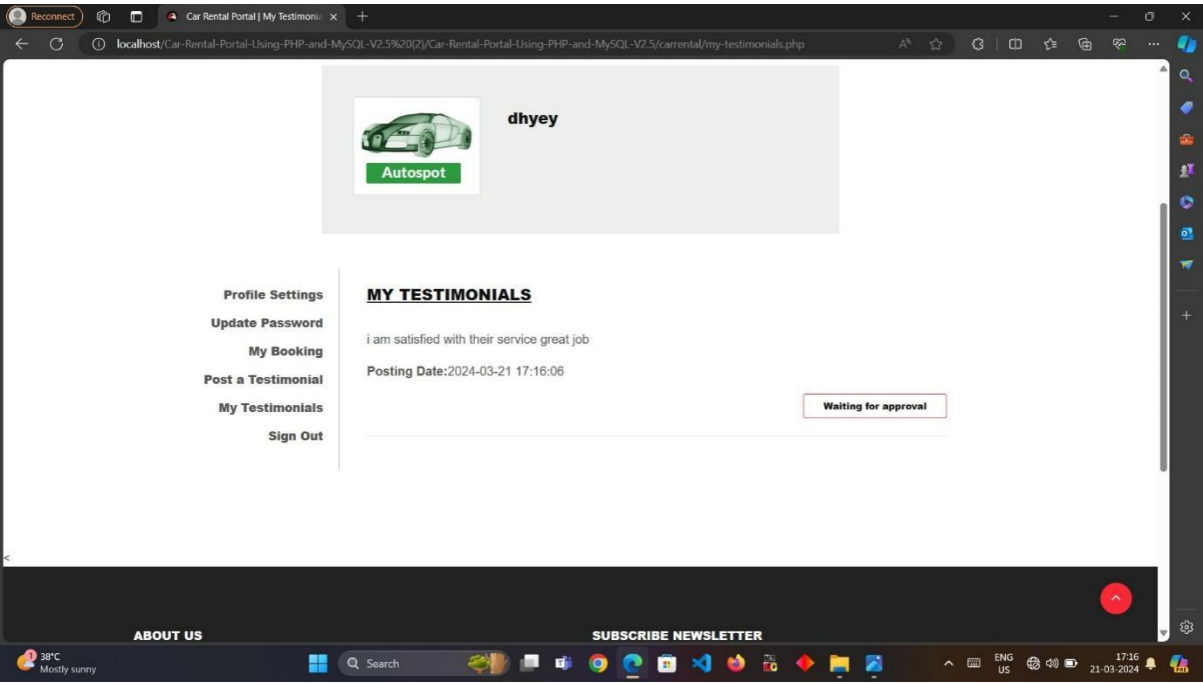
CAR RENTAL MANAGEMENT SYSTEM



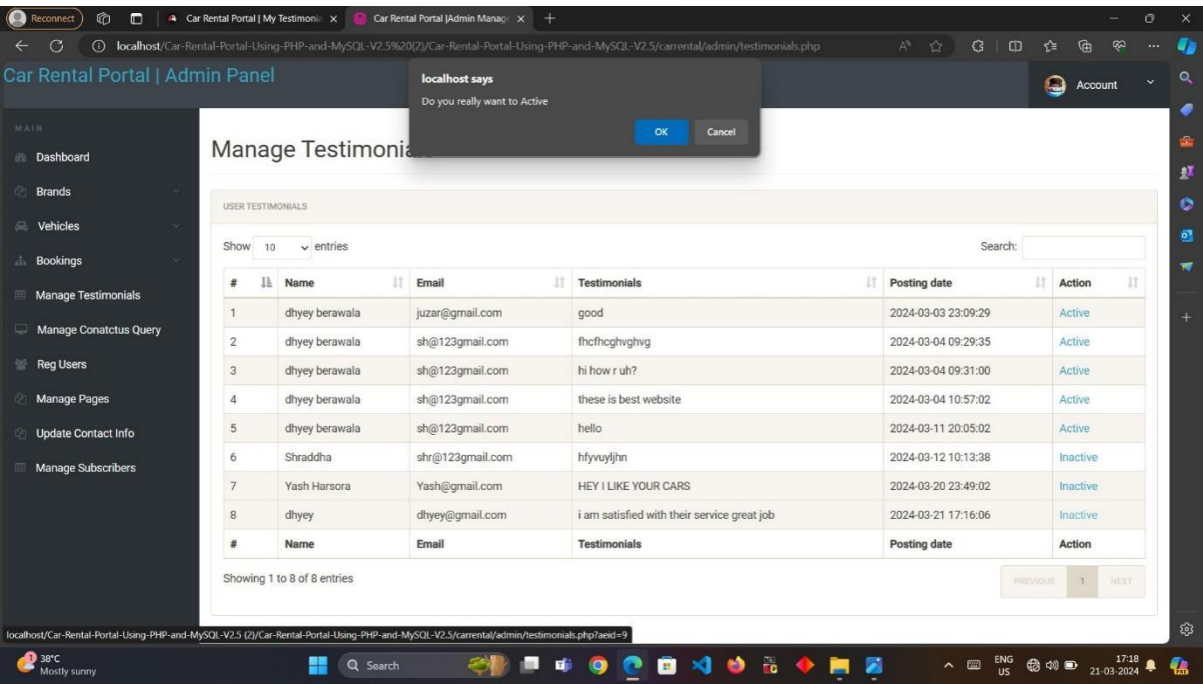
- User Testimonial Page: User can give feedback.



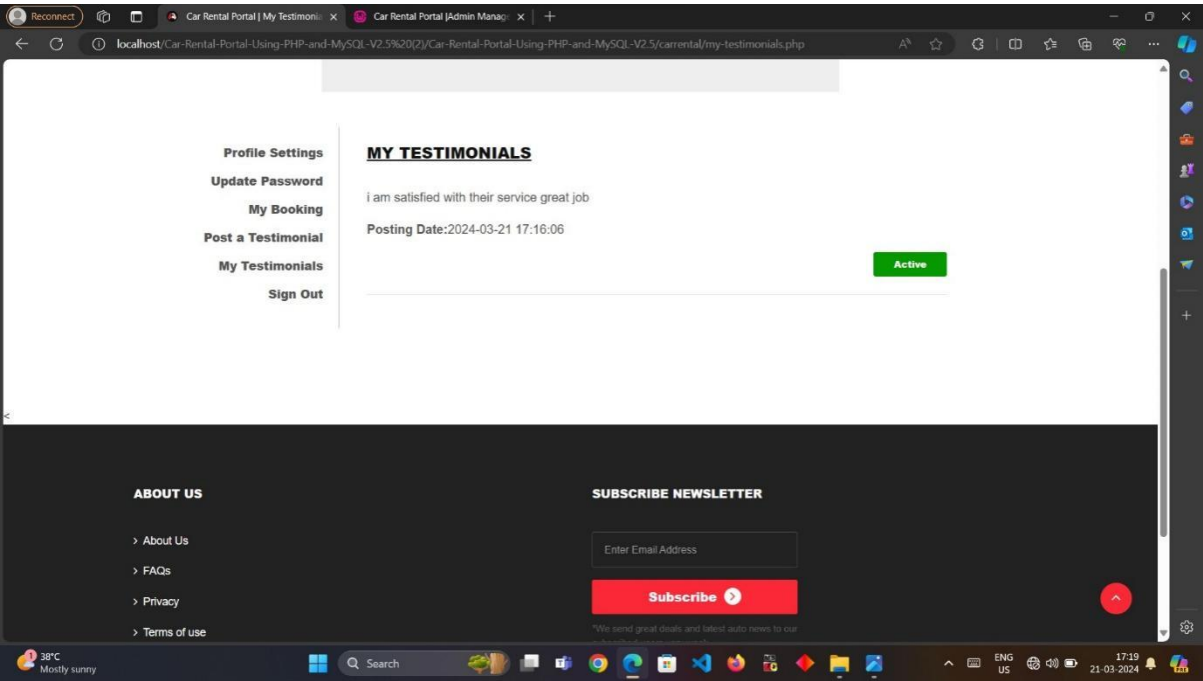
CAR RENTAL MANAGEMENT SYSTEM



- Admin Manage Testimonial Page: Admin can see user’s feedback.

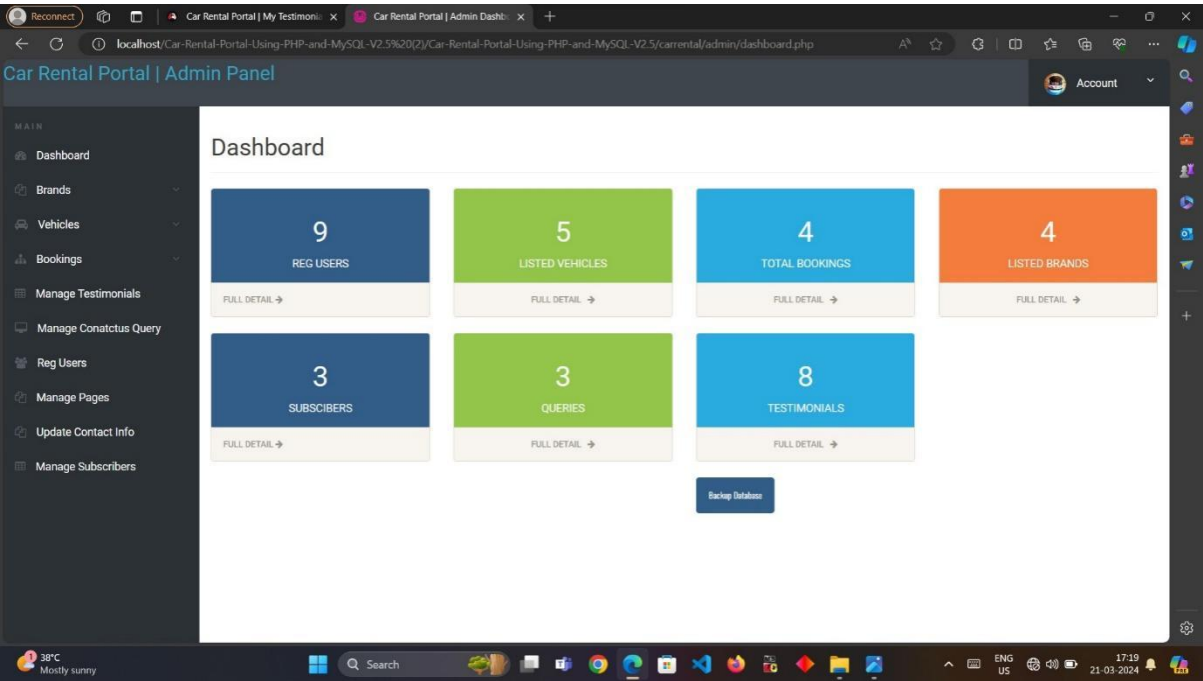


CAR RENTAL MANAGEMENT SYSTEM



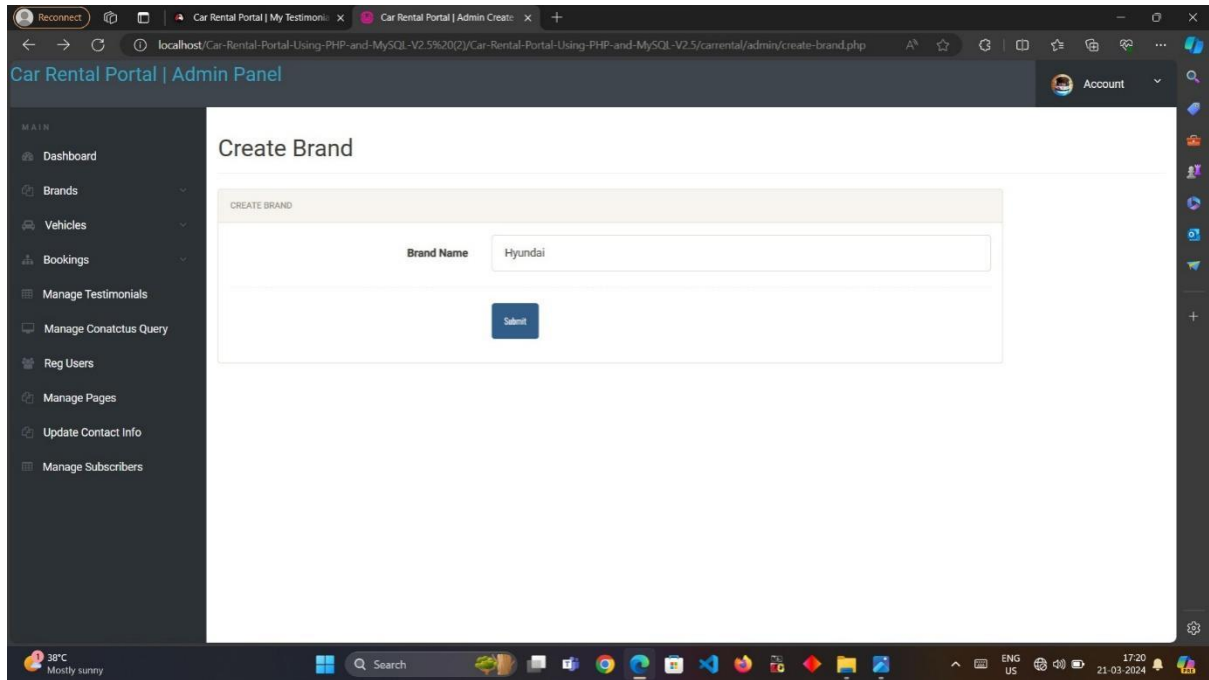
ADMIN SIDE

- Admin Dashboard: Admin side dashboard.

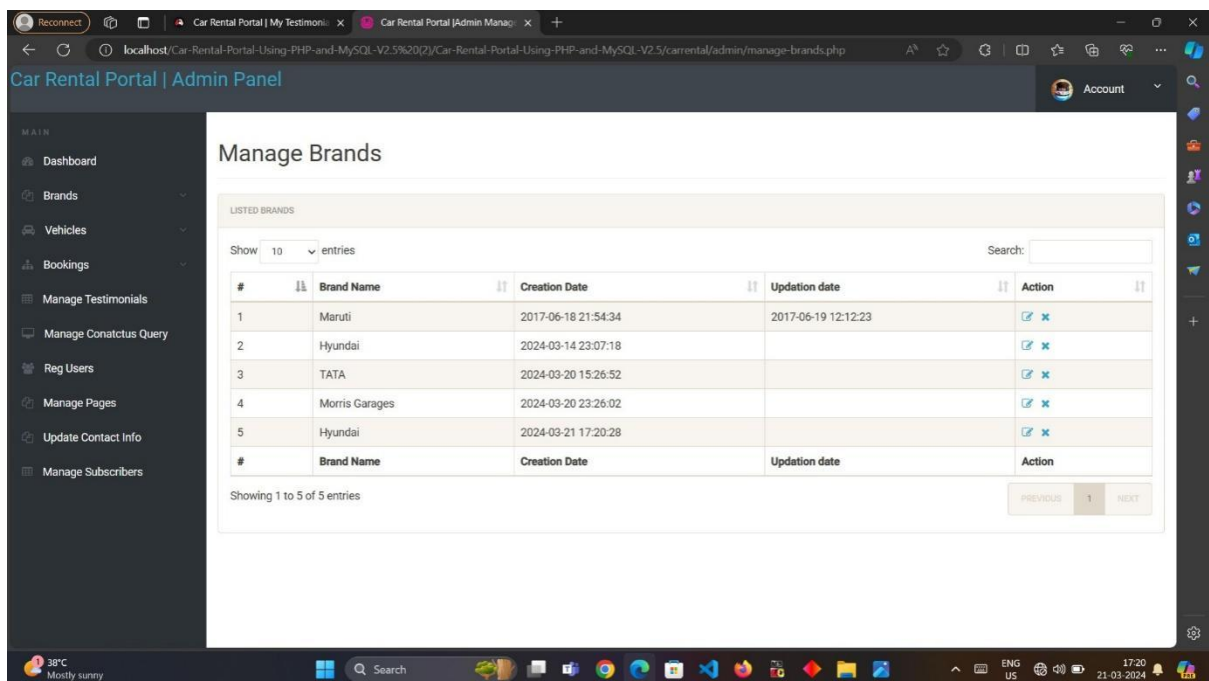


CAR RENTAL MANAGEMENT SYSTEM

- Admin Create Brand: Admin can create brand.



- Admin Manage Brands: Here admin can manage vehicle brands.



CAR RENTAL MANAGEMENT SYSTEM

- **Admin Add Vehicle:** Here admin can add vehicle.

Car Rental Portal | Admin Panel

Post A Vehicle

BASIC INFO

Vehicle Title* Select Brand* Hyundai

Vehical Overview*

Price Per Day(in USD)* Select Fuel Type* Petrol

Model Year* Seating Capacity*

Upload Images

Image 1* download.jpeg

Image 2* images (1).jpeg

Image 3* images (2).jpeg

Image 4* images (3).jpeg

Image 5 images.jpeg

Car Rental Portal | Admin Panel

Price Per Day(in USD)* Select Fuel Type* Petrol

Model Year* Seating Capacity*

Upload Images

Image 1* download.jpeg

Image 2* images (1).jpeg

Image 3* images (2).jpeg

Image 4* images (3).jpeg

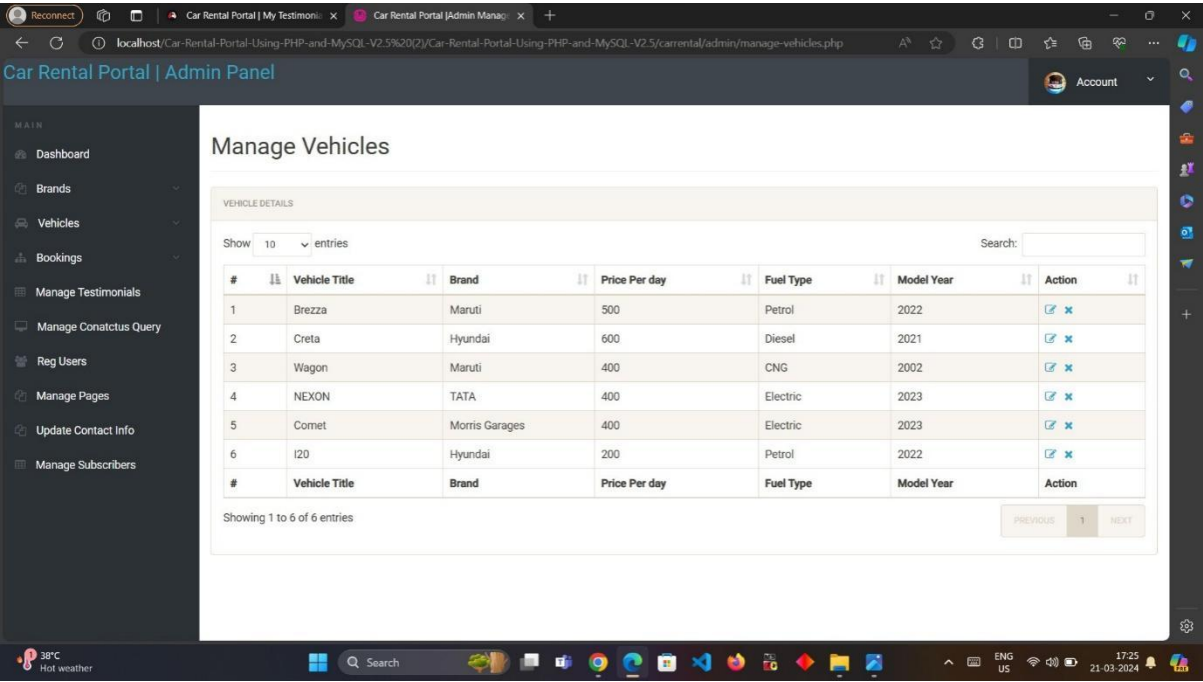
Image 5 images.jpeg

ACCESSORIES

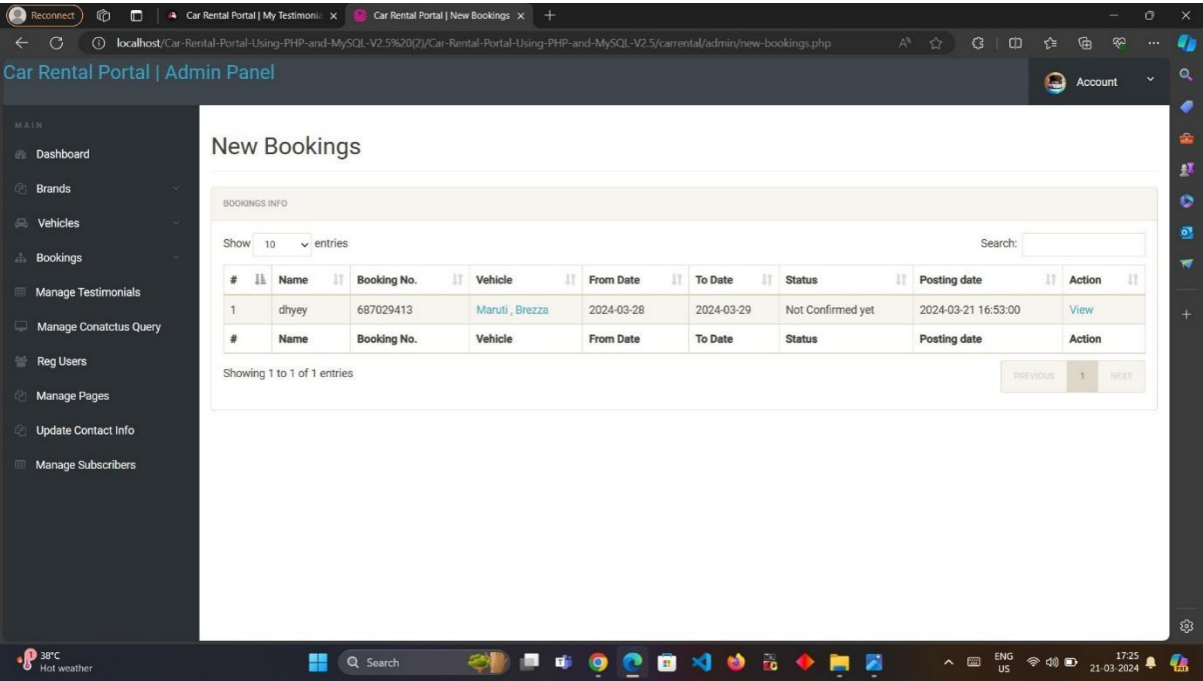
☒ Air Conditioner	☒ Power Door Locks	☒ AntiLock Braking System	☒ Brake Assist
☒ Power Steering	☒ Driver Airbag	☒ Passenger Airbag	☒ Power Windows
☒ CD Player	☒ Central Locking	☒ Crash Sensor	☒ Leather Seats

CAR RENTAL MANAGEMENT SYSTEM

- Admin Manage Vehicles: Admin can manage vehicles.



- Admin Booking: Here admin can see new bookings.



CAR RENTAL MANAGEMENT SYSTEM

Reconnect

Car Rental Portal | My Testimoni...

Car Rental Portal | Confirmed Bo...

+

localhost/Car-Rental-Portal-Using-PHP-and-MySQL-V2.5%20(2)/Car-Rental-Portal-Using-PHP-and-MySQL-V2.5/carrental/admin/confirmed-bookings.php

Car Rental Portal | Admin Panel

Account

MAIN

Dashboard

Brands

Vehicles

Bookings

Manage Testimonials

Manage Contactus Query

Reg Users

Manage Pages

Update Contact Info

Manage Subscribers

Confirmed Bookings

BOOKINGS INFO

Show 10 entries

Search:

#	Name	Booking No.	Vehicle	From Date	To Date	Status	Posting date	Action
1	dhyey	113942476	TATA , NEXON	2024-03-22	2024-03-25	Confirmed	2024-03-21 16:35:12	View
2	dhyey	576163440	Morris Garages , Comet	2024-04-05	2024-04-06	Confirmed	2024-03-21 17:04:11	View
#	Name	Booking No.	Vehicle	From Date	To Date	Status	Posting date	Action

Showing 1 to 2 of 2 entries

PREVIOUS

1

NEXT

38°C

Hot weather

Search

ENG US

17:25

21-03-2024

Reconnect

Car Rental Portal | My Testimoni...

Car Rental Portal | Canceled Boo...

+

localhost/Car-Rental-Portal-Using-PHP-and-MySQL-V2.5%20(2)/Car-Rental-Portal-Using-PHP-and-MySQL-V2.5/carrental/admin/canceled-bookings.php

Car Rental Portal | Admin Panel

Account

MAIN

Dashboard

Brands

Vehicles

Bookings

Manage Testimonials

Manage Contactus Query

Reg Users

Manage Pages

Update Contact Info

Manage Subscribers

Canceled Bookings

BOOKINGS INFO

Show 10 entries

Search:

#	Name	Booking No.	Vehicle	From Date	To Date	Status	Posting date	Action
1	dhyey	320360692	Hyundai , Creta	2024-03-22	2024-03-30	Cancelled	2024-03-21 16:42:36	View
#	Name	Booking No.	Vehicle	From Date	To Date	Status	Posting date	Action

Showing 1 to 1 of 1 entries

PREVIOUS

1

NEXT

38°C

Mostly sunny

Search

ENG US

17:26

21-03-2024

CAR RENTAL MANAGEMENT SYSTEM

- User Invoice: Here admin can user’s invoice.

Car Rental Portal | Admin Panel

BOOKINGS INFO

#687029413 Booking Details

User Details			
Booking No.	#687029413	Name	dhyey
Email Id	dhyey@gmail.com	Contact No	8780804891
Address		City	
Country			

Booking Details			
Vehicle Name	Maruti , Brezza	Booking Date	2024-03-21 16:53:00
From Date	2024-03-28	To Date	2024-03-29
Total Days	1	Rent Per Days	500
Grand Total			500
Booking Status	Not Confirmed yet	Last pdation Date	

Confirm Booking Cancel Booking

Print

Car Rental Portal | Admin Panel

Booking Details

BOOKINGS INFO

#113942476 Booking Details

User Details			
Booking No.	#113942476	Name	dhyey
Email Id	dhyey@gmail.com	Contact No	8780804891
Address		City	
Country			

Booking Details			
Vehicle Name	TATA , NEXON	Booking Date	2024-03-21 16:35:12
From Date	2024-03-22	To Date	2024-03-25
Total Days	3	Rent Per Days	400
Grand Total			1200
Booking Status	Confirmed	Last pdation Date	

Print

CAR RENTAL MANAGEMENT SYSTEM

- **Manage Contact Us Query Page:** Admin can manage the user query.

Car Rental Portal | Admin Panel

Manage Contact Us Queries

USER QUERIES

Show 10 entries Search:

#	Name	Email	Contact No	Message	Posting date	Action
1	dhyey	dhyeyberawala@gmail.com	8780804891	this car rental system is best	2024-03-02 12:34:02	Read
2	Dhruv	Dhruv@gmail.com	9723383348	hello	2024-03-11 19:59:29	Read
3	Shraddha	sh@123gmail.com	9887576557	gvjhtnb	2024-03-12 10:12:32	Read

Showing 1 to 3 of 3 entries

PREVIOUS 1 NEXT

- **User Registered Page:** It shows users registered details.

Car Rental Portal | Admin Panel

Registered Users

REG USERS

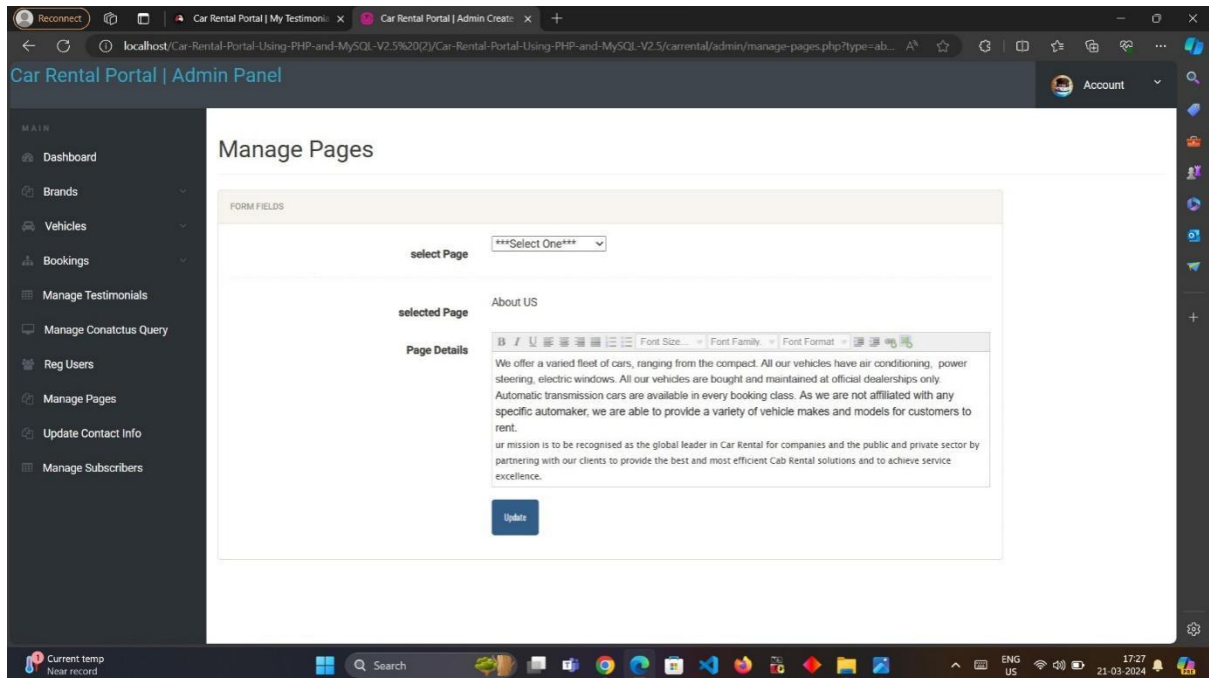
Show 10 entries Search:

#	Name	Email	Contact no	DOB	Address	City	Country	Reg Date
1	dhyey	dhyeyberawala@gmail.com	8780804891					2024-02-28 12:01:28
2	dhyey berawala	juzar@gmail.com	08780804891	29/01/2003	mangla park	surat	India	2024-03-03 18:43:19
3	dhyey berawala	sh@123gmail.com	08780804891	29/01/2003	mangla park	surat	India	2024-03-04 09:20:50
4	Shraddha	shr@123gmail.com	9887576557					2024-03-12 09:39:09
5	kruti	kruti@gmail.com	8866831120					2024-03-14 10:25:15
6	keval	keval@gmail.com	9586972070					2024-03-14 22:55:45
7	dhyey	cyberjuzar@gmail.com	8780804891					2024-03-16 09:50:12
8	dhyey	dhyey@gmail.com	8780804891					2024-03-19 12:53:44
9	Yash Harsora	Yash@gmail.com	7405142117	29/04/1995	shreeji krupa society , Bharuch	Bharuch	India	2024-03-20 23:42:51

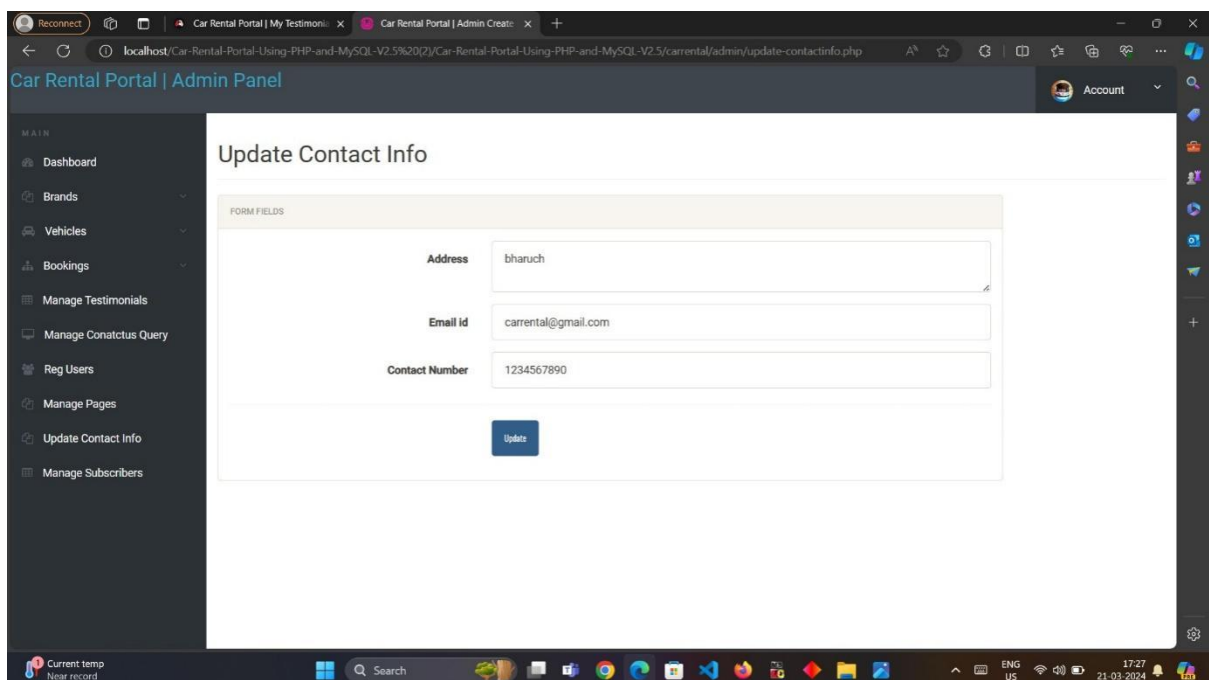
Showing 1 to 9 of 9 entries

PREVIOUS 1 NEXT

- **Manage Page:** Here admin can manage pages.

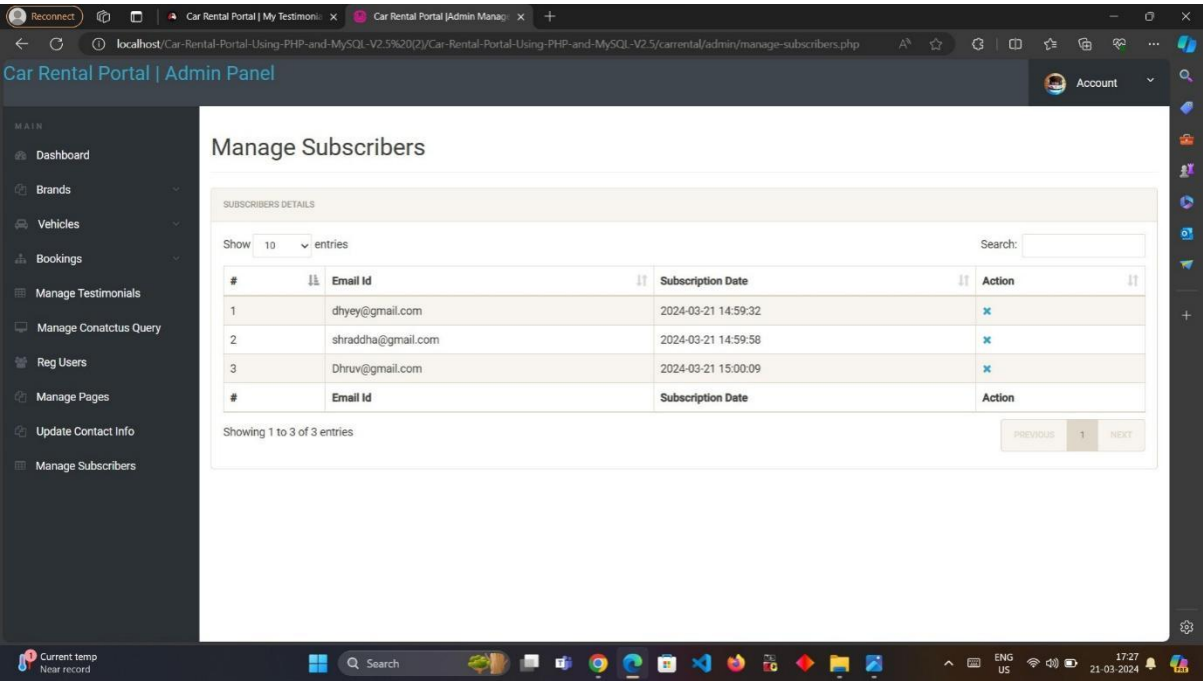


- **Update Contact Info:** Admin can update contact info.

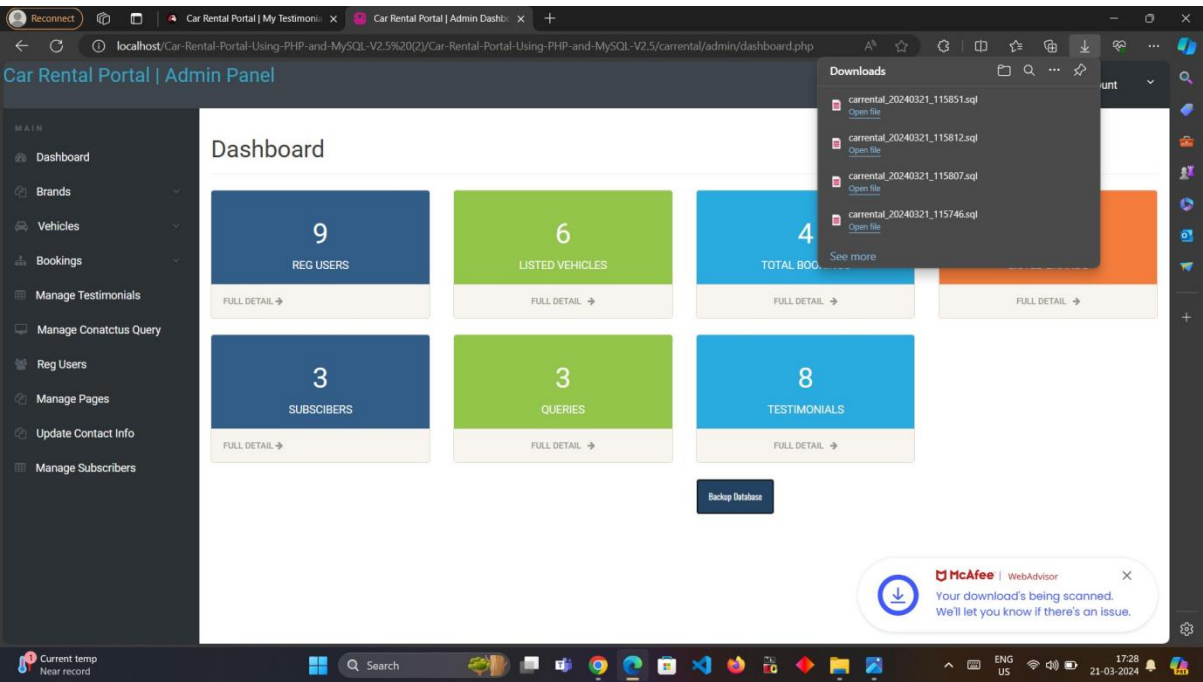


CAR RENTAL MANAGEMENT SYSTEM

- Manage Subscribers: Admin can manage subscribers.



- Backup And Restore



7.2 BACKUP AND RESTORE

Because data is the heart of enterprise, it's crucial for you to protect it. And to protect your organizations data you need to implement a data backup and recovery plan. Backing up files can protect against accidental loss of user data, database corruption, hardware failures and even natural disasters. It's your job as an administrator to make sure that backups are performed and that backup tapes are stored in a secure location.

8. SYSTEM TESTING

1. **Unit Testing:** Test individual components and modules of the system in isolation to ensure they function as expected. Verify that basic functionalities, such as user registration, reservation creation, and inventory management, work correctly.
2. **Functional Testing:** Validate that all specified functionalities work according to the requirements. Test scenarios such as vehicle reservation, rental agreement creation, payment processing, and return procedures.
3. **User Interface (UI) Testing:** Ensure that the user interface is user-friendly, intuitive, and follows the design specifications. Test the system's responsiveness across different devices and browsers.
4. **Performance Testing:** Assess the system's performance under various conditions, including peak usage. Measure response times for critical functions like reservation creation, payment processing, and search queries.
5. **Security Testing:** Identify and address potential security vulnerabilities. Test for data protection, encryption, and secure communication between the system and external services.
6. **Database Testing:** Validate data integrity, accuracy, and consistency within the database. Test database interactions, including data retrieval, storage, and updates.
7. **Load Testing:** Evaluate the system's ability to handle a large number of concurrent users. Assess how well the system scales to meet increased demand during peak times.
8. **Cross-Browser and Cross-Device Testing:** Verify that the system functions correctly across different browsers and devices. Test on various platforms to ensure a consistent user experience.
9. **Documentation Verification:** Confirm that all system documentation, including user manuals and technical documentation, is accurate and up-to-date.

9. FUTURE ENHANCEMENT

1. **Integration with Smart Technologies:** Incorporating features like GPS tracking, IoT sensors, and smart car technology to enhance vehicle monitoring and customer experience.
2. **Mobile App Enhancements:** Improving the mobile application for users to easily book, locate, and manage rentals on the go.
3. **Predictive Analytics:** Implementing predictive analytics to optimize fleet management, pricing strategies, and demand forecasting.
4. **Enhanced Customer Profiles:** Creating more detailed customer profiles to personalize services and promotions based on individual preferences and history.
5. **Blockchain for Transactions:** Exploring blockchain technology for secure and transparent rental transactions, ensuring data integrity and reducing fraud risks.
6. **Automated Maintenance Alerts:** Implementing a system that sends automated alerts for vehicle maintenance based on usage patterns and mileage, reducing downtime.
7. **Contactless Transactions:** Enhancing contactless options for check-in, check-out, and payment processes to adapt to changing customer preferences and safety measures.
8. **Environmental Sustainability:** Introducing features to promote environmentally friendly practices, such as electric vehicle rentals and carbon offset programs.
9. **Advanced Reporting and Analytics:** Providing more robust reporting tools and analytics to enable better decision-making for business owners.
10. **User Feedback Integration:** Enhancing the system with features that allow customers to provide feedback easily, helping improve overall service quality.

10. BIBLIOGRAPHY

WEBSITES

- [HTTPS://WWW.W3SCHOOLS.COM/](https://www.w3schools.com/)
- [HTTPS://FONTAWESOME.COM/](https://fontawesome.com/)
- [HTTPS://STACKOVERFLOW.COM/](https://stackoverflow.com/)
- [HTTPS://W3LAYOUTS.COM/](https://w3layouts.com/)

BOOKS

- PHP Manual
- PHP Programming
- Head First PHP & MySQL
- PHP & MySQL Web Development
- Learn PHP